VECTOR 3D

Vector 3D

1. If $\vec{a}$ is a nonzero vector such that its projections on the vectors $2\hat{i}-\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\vec{a}$ is :

   यदि एक शून्येत्तर सदिश $\vec{a}$ के सदिशों $2\hat{i}-\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-2\hat{k}$ और $\hat{k}$ पर प्रक्षेप बराबर हैं, तो $\vec{a}$ के अनुदिश एक मात्रक सदिश है :

   (1) $\frac{1}{\sqrt{155}}\left(-7\hat{i}+9\hat{j}+5\hat{k}\right)$ (2) $\frac{1}{\sqrt{155}}\left(7\hat{i}+9\hat{j}+5\hat{k}\right)$

   (3) $\frac{1}{\sqrt{155}}\left(-7\hat{i}+9\hat{j}-5\hat{k}\right)$ (4) $\frac{1}{\sqrt{155}}\left(7\hat{i}+9\hat{j}-5\hat{k}\right)$

Ans. B

Sol. Let $\overline{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$a_1^2+a_2^2+a_3^2=1$

Let $\overline{b}=2\vec{i}-\hat{j}+2\hat{k}, \overline{c}=\vec{i}-2\hat{j}-2\hat{k}$

$\overline{d}=\hat{k}$

$$\frac{\overline{a}.\overline{b}}{|b|}=\frac{\overline{a}.\overline{c}}{|c|}=\frac{\overline{a}.\overline{d}}{|d|}$$

$$\frac{2a_1-a_2+2a_3}{3}=\frac{a_1+2a_2-2a_3}{3}=a_3$$

By solving

$$a_1=\frac{7}{\sqrt{155}},\ a_2=\frac{9}{\sqrt{155}},\ a_3=\frac{5}{\sqrt{155}}$$

2. Let $\vec{a}=2\hat{i}-3\hat{j}+\hat{k}$, $\vec{b}=3\hat{i}+2\hat{j}+5\hat{k}$ and a vector $\vec{c}$ be such that $(\vec{a}-\vec{c})\times\vec{b}=-18\hat{i}-3\hat{j}+12\hat{k}$ and $\vec{a}\cdot\vec{c}=3$. If $\vec{b}\times\vec{c}=\vec{d}$, then $|\vec{a}\cdot\vec{d}|$ is equal to :

   माना $\vec{a}=2\hat{i}-3\hat{j}+\hat{k}$, $\vec{b}=3\hat{i}+2\hat{j}+5\hat{k}$ और एक सदिश $\vec{c}$ के लिए $(\vec{a}-\vec{c})\times\vec{b}=-18\hat{i}-3\hat{j}+12\hat{k}$ तथा $\vec{a}\cdot\vec{c}=3$ हैं। यदि $\vec{b}\times\vec{c}=\vec{d}$ है, तो $|\vec{a}\cdot\vec{d}|$ बराबर है :

   (1) 12 (2) 18 (3) 15 (4) 9

Ans. C

Sol. $(\vec{a}-\vec{c})\times\vec{b}=-18\hat{i}-3\hat{j}+12\hat{k}$

$\vec{a}\times\vec{b}-\vec{c}\times\vec{b}=-18\hat{i}-3\hat{j}+12\hat{k}$

$\Rightarrow \vec{a}\times\vec{b}+\vec{d}=-18\hat{i}-3\hat{j}+12\hat{k}$

Take dot with $\vec{a}$

$\vec{a}\cdot\vec{d}=-36+9+12=-15$

$\therefore \left|\vec{a}\cdot\vec{d}\right|=15$

3. The line L_1 is parallel to vector $\vec{a}=-3\hat{i}+2\hat{j}+4\hat{k}$ and passes through the point (7, 6, 2) and the line L_2 is parallel to the vector $\vec{b}=2\hat{i}+\hat{j}+3\hat{k}$ and passes trough the point (5, 3, 4). The shortest distance between the lines L_1 and L_2 is :

रेखा L_1 सदिश $\vec{a}=-3\hat{i}+2\hat{j}+4\hat{k}$ के समान्तर है और बिन्दु (7, 6, 2) से होकर जाती है तथा रेखा L_2 सदिश $\vec{b}=2\hat{i}+\hat{j}+3\hat{k}$ के समान्तर है और बिन्दु (5, 3, 4) से होकर जाती है। रेखाओं L_1 और L_2 के बीच न्यूनतम दूरी है :

(1) $\frac{23}{\sqrt{38}}$ (2) $\frac{21}{\sqrt{57}}$ (3) $\frac{23}{\sqrt{57}}$ (4) $\frac{21}{\sqrt{38}}$

Ans. A

Sol. Equation of L_1 : $7\hat{i}+6\hat{j}+2\hat{k}+\lambda\left(-3\hat{i}+2\hat{j}+4\hat{k}\right)$

Equation of L_2 : $5\hat{i}+3\hat{j}+4\hat{k}+\lambda\left(2\hat{i}+\hat{j}+3\hat{k}\right)$

Shortest distance between L_1 and L_2

$$=\left|\frac{\left(\vec{a}_2-\vec{a}_1\right)\cdot\left(\vec{b}_1\times\vec{b}_2\right)}{\left|\vec{b}_1\times\vec{b}_2\right|}\right|$$

$$=\left|\frac{\left(-2\hat{i}-3\hat{j}+2\hat{k}\right)\cdot\left(2\hat{i}+17\hat{j}-7\hat{k}\right)}{\sqrt{2^2+(-17)^2+(-7)^2}}\right|$$

$$=\left|\frac{-4-51-14}{\sqrt{4+289+49}}\right|$$

$$=\left|\frac{-69}{\sqrt{342}}\right|=\frac{23}{\sqrt{38}}$$

4. Consider two vectors $\vec{u}=3\hat{i}-\hat{j}$ and $\vec{v}=2\hat{i}+\hat{j}-\lambda\hat{k},\lambda>0$. The angle between them is given by $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$.

Let $\vec{v}=\overrightarrow{v_1}+\overrightarrow{v_2}$, where $\overrightarrow{v_1}$ is parallel to $\vec{u}$ and $\overrightarrow{v_2}$ is perpendicular to $\vec{u}$. Then the value $\left|\overrightarrow{v_1}\right|^2+\left|\overrightarrow{v_2}\right|^2$ is equal to

:

दो सदिशों $\vec{u} = 3\hat{i} - \hat{j}$ और $\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}, \lambda > 0$ का विचार कीजिए। इनके बीच का कोण $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$ है।

माना $\vec{v} = \overrightarrow{v_1} + \overrightarrow{v_2}$ है, जहाँ $\overrightarrow{v_1}$ सदिश $\vec{u}$ के समान्तर है और $\overrightarrow{v_2}$ सदिश $\vec{u}$ के लंबवत है। तो $\left|\overrightarrow{v_1}\right|^2 + \left|\overrightarrow{v_2}\right|^2$ का मान है :

(1) 14 (2) $\frac{25}{2}$ (3) $\frac{23}{2}$ (4) 10

Ans. A

Sol. $\vec{u} = 3\hat{i} - \hat{j}$

$\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}\,; \lambda > 0$

$$\cos\theta = \frac{6-1}{\sqrt{10}\sqrt{5+\lambda^2}} = \frac{\sqrt{5}}{2\sqrt{7}}$$

$$\frac{\sqrt{5}}{\sqrt{10}\sqrt{5+\lambda^2}} = \frac{\sqrt{5}}{\sqrt{2}\sqrt{7}}$$

$5 + \lambda^2 = 14$

$\lambda^2 = 14 - 5$

$\lambda = \sqrt{9}$

$\lambda = 3$

$\therefore \vec{v} = 2\hat{i} + \hat{j} - 3\hat{k}$

$\vec{v} = \overrightarrow{v_1} + \overrightarrow{v_2}$ $\left\{\begin{array}{l}\overrightarrow{v_1} \parallel \vec{u} \\ \therefore \overrightarrow{v_1} = \lambda\vec{u}\end{array}\right\}$

$= \alpha\left(3\hat{i} - \hat{j}\right) + \overrightarrow{v_2}$

$\Rightarrow \overrightarrow{v_2} = \vec{v} - \alpha\left(3\hat{i} - \hat{j}\right)$

$= 2i + j - 3k - 3\alpha i + \alpha j$

$\vec{v}_2 = (2 - 3\alpha)\hat{i} + (1 + \alpha)\hat{j} - 3\hat{k}$

Now, $\overrightarrow{v_2}$ is $\perp$r to $\vec{u}$

$\overrightarrow{v_2}.\vec{u} = 0$

$(2 - 3\alpha)3 + (1 + \alpha)(-1) = 0$

$6 - 9\alpha - 1 - \alpha = 0$

$5 = 10\alpha$

$\alpha = \frac{1}{2}$

$\overrightarrow{v_1} = \frac{3i}{2} - \frac{j}{2}$

$\left|\overrightarrow{v_1}\right|^2 = \frac{9}{4} + \frac{1}{4} = \frac{10}{4}$

$= \frac{5}{2}$

$\overrightarrow{v_2} = \left(2 - \frac{3}{2}\right)i + \left(1 + \frac{1}{2}\right)j - 3k$

$= \frac{1}{2}\hat{i} + \frac{3}{2}j - 3k$

$\left|\overrightarrow{v_2}\right| = \frac{1}{4} + \frac{9}{4} + 9$

$\Rightarrow \frac{10}{4} + 9$

$\Rightarrow \frac{5}{2} + 9$

$\left|\overrightarrow{v_1}\right|^2 + \left|\overrightarrow{v_2}\right|^2 = 5 + 9 = 14$

5. Let the angle $\theta, 0 < \theta < \frac{\pi}{2}$ between two unit vectors $\hat{a}$ and $\hat{b}$ be $\sin^{-1}\left(\frac{\sqrt{65}}{9}\right)$. If the vector $\vec{c} = 3\hat{a} + 6\hat{b} + 9(\hat{a} \times \hat{b})$, then the value of $9(\vec{c} \cdot \hat{a}) - 3(\vec{c} \cdot \hat{b})$ is :

माना दो मात्रक सदिशों $\hat{a}$ और $\hat{b}$ के बीच कोण $\theta = \sin^{-1}\left(\frac{\sqrt{65}}{9}\right), 0 < \theta < \frac{\pi}{2}$ है। यदि सदिश $\vec{c} = 3\hat{a} + 6\hat{b} + 9(\hat{a} \times \hat{b})$ है, तो $9(\vec{c} \cdot \hat{a}) - 3(\vec{c} \cdot \hat{b})$ का मान है :

(1) 29 (2) 27 (3) 31 (4) 24

Ans. A

Sol. $\vec{c} = 3\vec{a} + 6\vec{b} + 9\left(\vec{a} \times \vec{b}\right)$

$\sin^{-1}\left(\frac{\sqrt{65}}{9}\right) \Rightarrow \sin\theta = \frac{\sqrt{65}}{9} \Rightarrow \cos\theta = \frac{4}{9}$

$\vec{c}.\vec{a} = 3\left|\vec{a}\right|^2 + 6\vec{a}.\vec{b} = 3 + \frac{6.4}{9} = \frac{51}{9}$

$\vec{c}.\vec{a} = 3\vec{a}.\vec{b} + 6|\vec{b}|^2 = \frac{3.4}{9} + 6 = \frac{22}{3}$

$\therefore 9(\vec{c}.\vec{a}) - 3(\vec{c}.\vec{b}) = 51 - 22 = 29$

6. If the equation of the line passing through the point $\left(0, -\frac{1}{2}, 0\right)$ and perpendicular to the lines $\vec{r} = \lambda\left(\hat{i} + a\hat{j} + b\hat{k}\right)$ and $\vec{r} = \left(\hat{i} - \hat{j} - 6\hat{k}\right) + \mu\left(-b\hat{i} + a\hat{j} + 5\hat{k}\right)$ is $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$, then a + b + c + d is equal to :

यदि बिन्दु $\left(0, -\frac{1}{2}, 0\right)$ से होकर जाने वाली तथा रेखाओं $\vec{r} = \lambda\left(\hat{i} + a\hat{j} + b\hat{k}\right)$ और $\vec{r} = \left(\hat{i} - \hat{j} - 6\hat{k}\right) + \mu\left(-b\hat{i} + a\hat{j} + 5\hat{k}\right)$ के लम्बवत् रेखा का समीकरण $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$ है, तो a + b + c + d बराबर है :

(1) 14 (2) 10 (3) 13 (4) 12

Ans. A

Sol. $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$

$\left(0, -\frac{1}{2}, 0\right)$ will satisfy it

$\frac{1}{2} = \frac{7}{2d} = \frac{c}{4}$

$\Rightarrow d = 7, c = 2$

Line is perpendicular to other lines

So,

$(-2, d, -4)\cdot(1, a, b) = 0$

$\Rightarrow -2 + 7a - 4b = 0$ ...(1)

and

$(-2, d, -4)\cdot(-b, a, 5) = 0$

$\Rightarrow 2b + 7a - 20 = 0$ ...(2)

from (1) and (2)

a = 2, b = 3

so, a + b + c + d = 2 + 3 + 2 + 7 = 14

7. Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\frac{|\vec{a}+\vec{b}| + |\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}| - |\vec{a}-\vec{b}|} = \sqrt{2} + 1$. Then $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ is :

माना बराबर परिमाण के दो सदिश $\vec{a}$ और $\vec{b}$ इस प्रकार हैं कि $\frac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|}=\sqrt{2}+1$ है। तो $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ बराबर है :

(1) $2+4\sqrt{2}$ (2) $1+\sqrt{2}$ (3) $2+\sqrt{2}$ (4) $4+2\sqrt{2}$

Ans. C

Sol. Let $|\vec{a}|=|\vec{b}|=P$

and $\vec{a}\wedge\vec{b}=\theta$

$$|\vec{a}+\vec{b}|=\sqrt{2}P\sqrt{1+\cos\theta}=P\cdot 2\cos\frac{\theta}{2}$$

$$|\vec{a}-\vec{b}|=P\cdot 2\sin\frac{\theta}{2}$$

So, $\frac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|}=\frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}=\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}$

$$\Rightarrow \tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=\sqrt{2}+1$$

$$\Rightarrow \frac{\pi}{4}+\frac{\theta}{2}=\frac{3\pi}{8}\Rightarrow\theta=\frac{\pi}{4}$$

Now, $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}=\frac{4P^2\cos^2\frac{\theta}{2}}{P^2}$

$$=4\cdot\cos^2\frac{\pi}{8}$$

$$=2\left(1+\cos\frac{\pi}{4}\right)=2\left(1+\frac{1}{\sqrt{2}}\right)$$

$$=\sqrt{2}\left(\sqrt{2}+\right)1$$

$$=2+\sqrt{2}$$

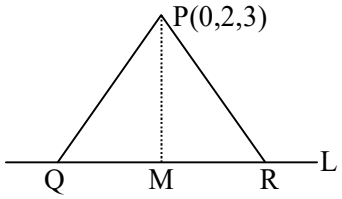
8. Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$, QR = 5 and the coordinates of the point P be (0, 2, 3). If the area of the triangle PQR is $\frac{m}{n}$:

माना त्रिभुज PQR के शीर्ष Q और R रेखा $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$ पर हैं, QR = 5 है और बिन्दु P के निर्देशांक (0, 2, 3) हैं। यदि त्रिभुज PQR का क्षेत्रफल $\frac{m}{n}$ है, तो :

(1) $2m-5\sqrt{21}n=0$ (2) $m-5\sqrt{21}n=0$ (3) $5m-2\sqrt{21}n=0$ (4) $5m-21\sqrt{2}n=0$

Ans. A

Sol.



QR = 5

M(5λ – 3, 2λ + 1, 3λ – 4)

Drs of PM $\Rightarrow$ 5λ – 3, 2λ – 1, 3λ – 7

Drs of line L $\Rightarrow$ 5, 2, 3

PM $\perp$ L

$\Rightarrow (5\lambda-3)5+(2\lambda-1)2+(3\lambda-7)3=0$

$\Rightarrow \lambda=1$

$\therefore$ M(2, 3, –1)

PM = $\sqrt{4+1+16}=\sqrt{21}$

Area = $\frac{1}{2}\times5\times\sqrt{21}=\frac{m}{n}$

$2m-5\sqrt{21}n=10$

9. Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the areas of the triangles ABC, ACD and ADB be 5, 6 and 7 square units respectively. Then the area (in square units) of the ΔBCD is equal to :

माना ABCD एक चतुष्फलक है जिसके किनारे AB, AC और AD परस्पर लंबवत हैं। माना त्रिभुजों ABC, ACD और ADB के क्षेत्रफल क्रमशः 5, 6 और 7 वर्ग इकाई हैं। तो त्रिभुज BCD का क्षेत्रफल (वर्ग इकाई में) बराबर है :

(1) $7\sqrt{3}$ (2) $\sqrt{110}$ (3) $\sqrt{340}$ (4) 12

Ans. B

Sol. Ar(ΔBCD)

$$=\sqrt{\left(\text{Ar}\left(\Delta ABC\right)\right)^2+\left(\text{Ar}\left(ACD\right)\right)^2+\left(\text{Ar}\left(\Delta ADB\right)\right)^2}$$

$= \sqrt{5^2 + 6^2 + 7^7}$

$= \sqrt{110}$

10. Line L_1 passes through the point (1, 2, 3) and is parallel to z-axis. Line L_2 passes through the point (λ, 5, 6) and is parallel to y-axis. Let for $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, the shortest distance between the two lines be 3. Then the square of the distance of the point (λ_1, λ_2, 7) from the line L_1 is :

रेखा L_1, बिन्दु (1, 2, 3) से होकर जाती है और z-अक्ष के समांतर है। रेखा L_2, बिन्दु (λ, 5, 6) से होकर जाती है और y-अक्ष के समांतर है। माना $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, के लिए इन दो रेखाओं के बीच न्यूनतम दूरी 3 है। तो बिन्दु (λ_1, λ_2, 7) की रेखा L_1 से दूरी का वर्ग है :

(1) 32 (2) 40 (3) 37 (4) 25

Ans. D

Sol. $L_1 : \frac{x-1}{0} = \frac{y-2}{0} = \frac{z-3}{1}$

$L_2 : \frac{x-\lambda}{0} = \frac{y-5}{1} = \frac{z-6}{0}$

S.D. $= \left| \frac{\overline{AB}(\bar{p} \times \bar{q})}{(\bar{p} \times \bar{q})} \right| = |\lambda - 1| = 3$

$\lambda = 4$ or $\lambda = -2$

Let foot of perpendicular from

P(4, –2, 7) is Q(1, 2, t+3)

(3, –4, 4–t).(0, 0, 1) = 0

t = 4

$PQ^2 = 25$

11. Let a line passing through the point (4, 1, 0) intersect the line $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at the point A(α, β, γ) and the line $L_2 : x - 6 = y = -z + 4$ at the point B(a, b, c). Then $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ is equal to :

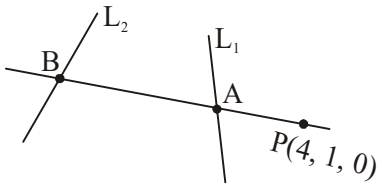
माना बिन्दु (4, 1, 0) से होकर जाने वाली एक रेखा, रेखा $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ को बिन्दु A(α, β, γ) और रेखा $L_2 : x - 6$

= y = –z + 4 को बिन्दु B(a, b, c) पर काटती है। तो $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ बराबर है :

(1) 6 (2) 8 (3) 16 (4) 12

Ans. B

Sol.



$L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = p$ A(2p + 1, 3p + 2, 4p + 3)

$L_2 : \frac{x-6}{1} = \frac{y}{1} = \frac{z-4}{-1} = q$ B(q + 6, q, 4 – q)

D.R. of PA = (2p – 3, 3p + 1, 4p + 3)
D.R. of PB = (q + 2, q – 1, 4 – q)

$$\frac{2p-3}{q+2} = \frac{3p+1}{q-1} = \frac{4p+3}{4-q}$$

2pq – 2p – 3q + 3 = 3pq + 6p + q + z
pq + rp + 4q – 1 = 0 .....(1)
12p – 3pq + 4 – q = 4pq + 3q – 4p – 3
7pq – 16p + 4q – 7 = 0 .....(2)
pq = – 3
p = –1 q = 3
A(–1, –1, –1) B(9, 3, 1)

$$\begin{vmatrix} 1 & 0 & 1 \\ -1 & -1 & -1 \\ 9 & 3 & 1 \end{vmatrix} = 8$$

12. Each of angles β and γ that a given line makes the positive y- and z-axes, respectively, is half of the angle that this line makes with the positive x-axes. Then the sum of all possible values of the angle β is :

दी गई एक रेखा के धनात्मक y- और z-अक्षों से क्रमशः बनाए गए कोणों β और γ में प्रत्येक, इस रेखा के धनात्मक x-अक्ष से बनाए गए कोण का आधा है। तो β के सभी संभव मानों का योग है :

(1) $\frac{3\pi}{4}$ (2) π (3) $\frac{3\pi}{2}$ (4) $\frac{\pi}{2}$

Ans. A

Sol. $\beta = \frac{\alpha}{2}, \gamma = \frac{\alpha}{2}$

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\cos^2\alpha + 2\cos^2\frac{\alpha}{2} = 1$

$\cos^2\alpha + \cos\alpha = 0$

$\cos\alpha(\cos\alpha + 1) = 0$

$\cos\alpha = 0, -1$

$\alpha = \frac{\pi}{2}, \pi$

Now, $\beta = \frac{\alpha}{2} \Rightarrow \frac{\pi}{4}, \frac{\pi}{2}$

So, sum is $\frac{3\pi}{4}$

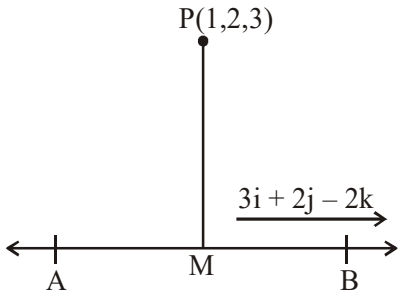
13. Let A and B be two distinct points on the line $L: \frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Both A and B are at a distance $2\sqrt{17}$ from the foot of perpendicular drawn from the point (1, 2, 3) on the line L. If O is the origin, then $\overrightarrow{OA} \cdot \overrightarrow{OB}$ is equal to :

माना रेखा $L: \frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ पर दो भिन्न बिन्दु A और B हैं। बिन्दु (1, 2, 3) से रेखा L पर डाले गए लंब के पाद से A और B दोनों $2\sqrt{17}$ की दूरी पर है। यदि O मूलबिन्दु है, तो $\overrightarrow{OA} \cdot \overrightarrow{OB}$ बराबर है :

(1) 62 (2) 21 (3) 49 (4) 47

Ans. D

Sol. $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2} = \lambda$



Let M be foot of perpendicular

M $\{3\lambda + 6, 2\lambda + 7, -2\lambda + 7\}$

$\overrightarrow{MP} = \{3\lambda + 5, 2\lambda + 5, -2\lambda + 4\}$

$\vec{b} = 3i + 2j - 2k$ (Direction ratio of line)

$\overrightarrow{MP}\cdot\vec{b} = 0$

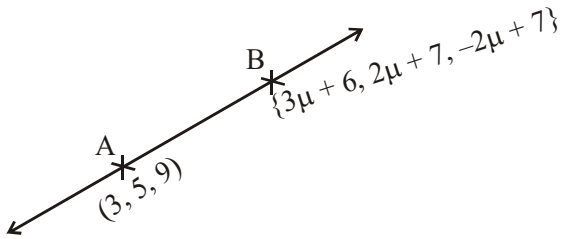
$(3\lambda + 5)\cdot 3 + (2\lambda + 5)\cdot 2 + (-2\lambda + 4)(-2) = 0$

$9\lambda + 15 + 4\lambda + 10 + 4\lambda - 8 = 0$

$17\lambda = -17$

$\lambda = -1$

$\therefore$ M {3, 5, 9}



A.T.Q.

$|AB| = 2\sqrt{17}$

$\sqrt{(3\mu+3)^2 + (2\mu+2)^2 + (-2\mu-2)^2} = \sqrt{17\times 4}$

$9(\mu + 1)^2 + 4(\mu + 1)^2 + 4(\mu + 1)^2 = 17 \times 4$

$17(\mu + 1)^2 = 17 \times 4$

$\mu + 1 = \pm 2$

$\mu = -1 \pm 2$

$\mu = 1, -3$

$\therefore$ B {9,9,5} or {–3, 1, 13}

$\overrightarrow{OA}\cdot\overrightarrow{OB} = 9 \times (-3) + 9 \times 1 + 5 \times 13$

$= -27 + 9 + 65$

$= 47$

14. Let A be the point of intersection of the lines $L_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ and $L_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$. Let B and C be the points on the lines L_1 and L_2 respectively such that AB = AC = $\sqrt{15}$. Then the square of the area of the triangle ABC is :

माना रेखाओं $L_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ और $L_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$ का प्रतिच्छेदन बिन्दु A है। माना रेखाओं L_1 और L_2 पर क्रमशः बिन्दु B और C इस प्रकार हैं कि AB = AC = $\sqrt{15}$ है। तो त्रिभुज ABC के क्षेत्रफल का वर्ग है :

(1) 57 (2) 63 (3) 60 (4) 54

Ans. D

Sol. $\text{Area} = \frac{1}{2}ab\sin\theta$

$$\Rightarrow \cos\theta = \left|\frac{-2}{\left(\sqrt{2}\right)\sqrt{50}}\right| = \frac{1}{5}$$

$$\sin\theta = \frac{\sqrt{24}}{5}$$

$$\text{Area} = \frac{1}{2}\left(\sqrt{15}\right)\left(\sqrt{15}\right)\left(\frac{\sqrt{24}}{5}\right) = \frac{3}{2}\sqrt{24}$$

$(\text{Area})^2 = 54$

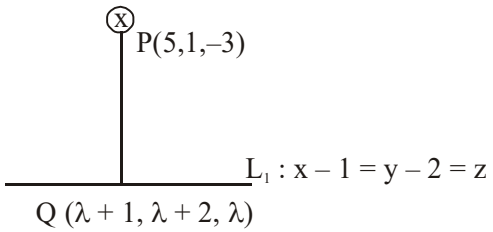
15. Consider the lines $L_1 : x-1 = y-2 = z$ and $L_2 : x-2 = y = z-1$. Let the feet of the perpendiculars from the point P(5, 1, –3) on the lines L_1 and L_2 be Q and R respectively. If the area of the triangle PQR is A, then $4A^2$ is equal to :

रेखाओं $L_1 : x-1 = y-2 = z$ और $L_2 : x-2 = y = z-1$ का विचार कीजिए। माना बिन्दु P(5, 1, –3) से रेखाओं L_1 और L_2 पर डाले गए लम्ब के पाद क्रमशः Q और R हैं। यदि त्रिभुज PQR का क्षेत्रफल A है, तो $4A^2$ बराबर है :

(1) 151 (2) 143 (3) 139 (4) 147

Ans. D

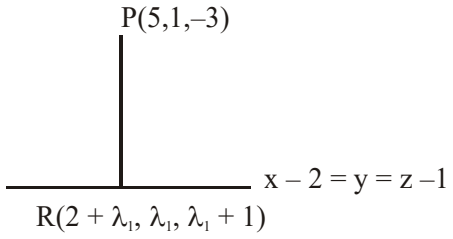
Sol.



$\overrightarrow{PQ}\cdot(1,1,1) = 0 \Rightarrow (\lambda-4)+(\lambda+1)+(\lambda+3) = 0$

$\Rightarrow 3\lambda = 0 \Rightarrow \lambda = 0$

So, point Q is (1, 2, 0)

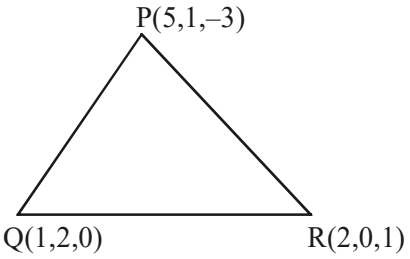


$\overrightarrow{PR}\cdot(1,1,1) = 0$

$\Rightarrow (\lambda_1-3, \lambda_1-1, \lambda_1+4)\cdot(1,1,1) = 0$

$\Rightarrow 3\lambda_1 = 0 \Rightarrow \lambda_1 = 0$

So, point R is (2, 0, 1)



$\overrightarrow{PQ} = (-4,1,3)$

$\overrightarrow{PR} = (-3,-1,4)$

$\overrightarrow{PQ} \times \overrightarrow{PR} = (7,7,7)$

so, area (A) = $\frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{7\sqrt{3}}{2}$

$4A^2 = \frac{4 \times 49 \times 3}{4} = 147$

16. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{3}$. If $\lambda\vec{a} + 2\vec{b}$ and $3\vec{a} - \lambda\vec{b}$ are perpendicular to each other, then the number of values of λ in [–1, 3] is :

माना कि $\vec{a}$ और $\vec{b}$ दो इकाई सदिश इस प्रकार हैं कि उनके बीच का कोण $\frac{\pi}{3}$ है। यदि $\lambda\vec{a} + 2\vec{b}$ और $3\vec{a} - \lambda\vec{b}$ एक दूसरे के लम्बवत् हैं, तो [–1, 3] में λ के मानों की संख्या है :

(1) 1 (2) 2 (3) 3 (4) 0

Ans. D

Sol. $|\vec{a}| = 1$ $|\vec{b}| = 1$

$\vec{a}.\vec{b} = |\vec{a}||\vec{b}|\cos\frac{\pi}{3} = \frac{1}{2}$

So $\left(\lambda\vec{a} + 2\vec{b}\right).\left(3\vec{a} - \lambda\vec{b}\right) = 0$

$3\lambda - \lambda^2 + 6\vec{a}.\vec{b} - \lambda\vec{a}.\vec{b} = 0$

$3\lambda - \lambda^2 + \frac{6}{2} - \frac{\lambda}{2} = 0$

$2\lambda^2 - 5\lambda + 6 = 0$

$D = 25 - 48 < 0$ No real value of λ

17. Let a line pass through two distinct points P(–2, –1, 3) and Q, and be parallel to the vector $3\hat{i} + 2\hat{j} + 2\hat{k}$. If the

distance of the point Q from the point R(1, 3, 3) is 5, then the square of the area of ΔPQR is equal to :

माना कि एक रेखा दो भिन्न बिन्दुओं P(–2, –1, 3) और Q से होकर गुजरती है और यह सदिश $3\hat{i}+2\hat{j}+2\hat{k}$ के समान्तर है। यदि बिन्दु Q की बिन्दु R(1, 3, 3) से दूरी 5 है, तो ΔPQR के क्षेत्रफल का वर्ग बराबर है :

(1) 148 (2) 140 (3) 136 (4) 144

Ans. C

Sol. P(–2, –1, 3)

equation of line PQ $\vec{r}=(-2,-1,3)+\lambda(3,2,2)$

Let $Q(-2+3\lambda,-1+2\lambda,3+2\lambda)$

R(1, 3, 3)

$QR=5=\sqrt{(-3+3\lambda)^2+(-4+2\lambda)^2+(2\lambda)^2}$

$25 = 17\lambda^2 - 34\lambda + 25$

$17\lambda^2 - 34\lambda = 0$

$\lambda = 0$ or $\lambda = 2$

Q(4, 3, 7)

Now Area of $\Delta PQR = \frac{1}{2}\left|\overrightarrow{PQ}\times\overrightarrow{QR}\right|$

$\overrightarrow{PQ}=6\hat{i}+4\hat{j}+4\hat{k}$

$\overrightarrow{QR}=-3\hat{i}+0\hat{j}-4\hat{k}$

$$\overrightarrow{PQ}\times\overrightarrow{QR}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\6&4&4\\-3&0&-4\end{vmatrix}=-16\hat{i}+12\hat{j}+12\hat{k}$$

$$\Delta=\frac{1}{2}\sqrt{256+144+144}=\frac{\sqrt{544}}{2}$$

$$\Delta^2=\frac{544}{4}=136$$

18. Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+3\hat{j}-2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median line segment through A of the triangle ABC at the point E. If the length of AD is $\frac{\sqrt{110}}{3}$ and the volume of the tetrahedron is $\frac{\sqrt{805}}{6\sqrt{2}}$, then the position vector of E is :

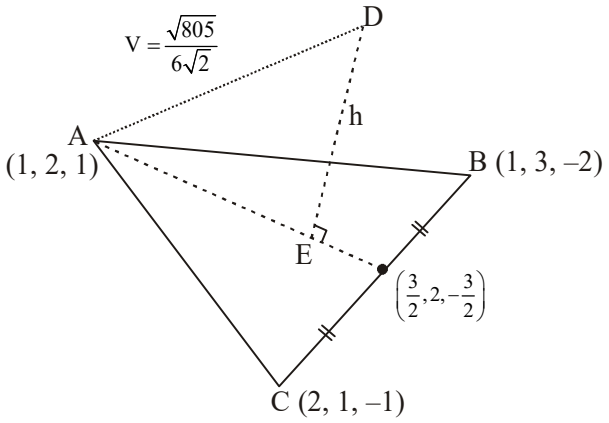
माना कि एक चतुष्फलक ABCD के शीर्षों A, B और C के स्थिति सदिश क्रमशः $\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+3\hat{j}-2\hat{k}$ और $2\hat{i}+\hat{j}-\hat{k}$ हैं। शीर्ष D से सम्मुख फलक ABC पर शीर्षलम्ब त्रिभुज ABC के माध्यिका रेखाखण्ड से, A के माध्यम से, बिन्दु E पर मिलता है। यदि AD की

लम्बाई $\frac{\sqrt{110}}{3}$ है और चतुष्फलक का आयतन $\frac{\sqrt{805}}{6\sqrt{2}}$ है, कतो E का स्थिति सदिश है :

(1) $\frac{1}{6}\left(12\hat{i}+12\hat{j}+\hat{k}\right)$ (2) $\frac{1}{2}\left(\hat{i}+4\hat{j}+7\hat{k}\right)$ (3) $\frac{1}{6}\left(7\hat{i}+12\hat{j}+\hat{k}\right)$ (4) $\frac{1}{12}\left(7\hat{i}+4\hat{j}+3\hat{k}\right)$

Ans. C

Sol.



Area of ΔABC

d.r. of $AE = \left(\frac{1}{2}, 0, -\frac{5}{2}\right) = (1, 0, -5)$

$\overline{AB} = \hat{j} - 3\hat{k}$

$\overline{AC} = \hat{i} - \hat{j} - 2\hat{k}$

$$\overline{AB} \times \overline{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & -3 \\ 1 & -1 & -2 \end{vmatrix}$$

$= -5\hat{i} - 3\hat{j} - \hat{k}$

Area of $\Delta ABC = \frac{1}{2}\sqrt{35}$

Area of tetrahedron $= \frac{1}{3}\left(\frac{1}{2}\sqrt{35}\right)h = \frac{\sqrt{805}}{6\sqrt{2}}$

$h = \sqrt{\frac{23}{2}}$

$AE = \sqrt{\frac{110}{9} - \frac{23}{2}} = \sqrt{\frac{220-207}{18}} = \sqrt{\frac{13}{18}}$

Equation of AE

$$\frac{x-1}{1}=\frac{y-2}{0}=\frac{z-1}{-5}=\lambda$$

Let $E(\lambda+1, 2, -5\lambda+1)$ $\qquad$ $A(1, 2, 1)$

$$AE=\sqrt{\frac{13}{18}}$$

$$\lambda^2+25\lambda^2=\frac{13}{18} \qquad E\left(\frac{7}{6},2,\frac{1}{6}\right)$$

$$26\lambda^2=\frac{13}{18}$$

$$\lambda=\pm\frac{1}{6}$$

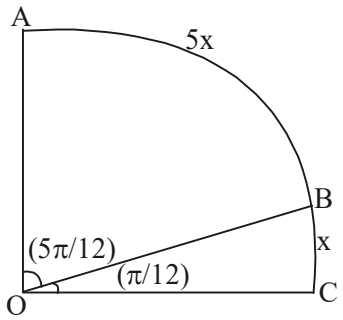
19. Let the arc AC of a circle subtend a right angle at the centre O. If the point B on the arc AC,divides the arc AC such that $\frac{\text{length of arc AB}}{\text{length of arc BC}}=\frac{1}{5}$, and $\overrightarrow{OC}=\alpha\overrightarrow{OA}+\beta\overrightarrow{OB}$, then $\alpha+\sqrt{2}\left(\sqrt{3}-1\right)\beta$ is equal to :

माना कि एक वृत्त का चाप AC, उसके केन्द्र O पर एक समकोण अंतरित करता है। यदि चाप AC पर बिन्दु B, चाप AC को इस प्रकार विभाजित करता है कि $\frac{\text{चाप AB की लम्बाई}}{\text{चाप BC की लम्बाई}}=\frac{1}{5}$ और $\overrightarrow{OC}=\alpha\overrightarrow{OA}+\beta\overrightarrow{OB}$ है, तो $\alpha+\sqrt{2}\left(\sqrt{3}-1\right)\beta$ बराबर है :

(1) $2+\sqrt{3}$ (2) $2-\sqrt{3}$ (3) $5\sqrt{3}$ (4) $2\sqrt{3}$

Ans. B

Sol.



$OC = OA = OB = r$

$$6x=\frac{\pi}{2}$$

$$x=\frac{\pi}{12}$$

$OC = OA = OB = r$

$\overrightarrow{OC}=\alpha\overrightarrow{OA}+\beta\overrightarrow{OB}$

dot with $\overrightarrow{OA}$

$0 = \alpha\left|\overrightarrow{OA}\right|^2 + \beta\overrightarrow{OA}\cdot\overrightarrow{OB}$

$0 = r^2\alpha + r^2\beta\cos\frac{\pi}{12}$

$\alpha + \frac{\sqrt{3}+1}{2\sqrt{2}}\beta = 0$ ...(1)

$\overrightarrow{OC} = \alpha\overrightarrow{OA} + \beta\overrightarrow{OB}$

dot with $\overrightarrow{OC}$

$r^2 = \alpha r^2\cos\frac{\pi}{12} + \beta r^2\cos\frac{5\pi}{12}$

$\beta\frac{\sqrt{3}-1}{2\sqrt{2}} = 1$ $\left(\sqrt{3}-1\right)\beta = 2\sqrt{2}$

$\beta = \sqrt{2}\left(\sqrt{3}+1\right)$

$\alpha + \frac{\sqrt{2}\left(\sqrt{3}+1\right)^2}{2\sqrt{2}} = 0$ $\alpha + \sqrt{2}\left(\sqrt{3}-1\right)\beta$

$\alpha = -\frac{3+1+2\sqrt{3}}{2}$ $= -2-\sqrt{3}+4$

$\alpha = -2-\sqrt{3}$ $= 2-\sqrt{3}$

20. Let the point A divide the line segment joining the points P(–1, –1, 2) and Q(5, 5, 10) internally in the ratio r : 1(r > 0). If O is the origin and $\left(\overrightarrow{OQ}\cdot\overrightarrow{OA}\right) - \frac{1}{5}\left|\overrightarrow{OP}\times\overrightarrow{OA}\right|^2 = 10$, then the value of r is :

माना कि बिन्दु A, बिन्दुओं P(–1, –1, 2) और Q(5, 5, 10) को मिलाने वाले रेखाखण्ड को अन्तः r : 1(r > 0) के अनुपात में विभाजित करता है। यदि O मूल बिन्दु है और $\left(\overrightarrow{OQ}\cdot\overrightarrow{OA}\right) - \frac{1}{5}\left|\overrightarrow{OP}\times\overrightarrow{OA}\right|^2 = 10$ है, तो r का मान है :

(1) 3 (2) 7 (3) $\sqrt{7}$ (4) 14

Ans. B

Sol. $A\left[\frac{5r-1}{r+1}, \frac{5r-1}{r+1}, \frac{10r+2}{r+1}\right]$

$\left(\overrightarrow{OQ}\cdot\overrightarrow{OA}\right) - \frac{\left|\overrightarrow{OP}\times\overrightarrow{OA}\right|^2}{5} = 10$ ...(1)

$\left(\overrightarrow{OQ}\cdot\overrightarrow{OA}\right)=\frac{5}{r+1}(30r+2)$

$\left|\overrightarrow{OP}\times\overrightarrow{OA}\right|^2=\frac{r^2}{(r+1)^2}(800)$

So, by equation (1)

$\frac{10}{r+1}(15r+1)-\frac{r^2(800)}{5(r+1)^2}=10$

$2r^2 - 14r = 0$

$r = 7, r \neq 0$

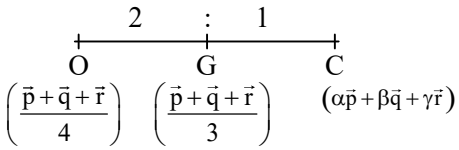
21. Let the position vectors of three vertices of a triangle be $4\vec{p}+\vec{q}-3\vec{r}, -5\vec{p}+\vec{q}+2\vec{r}$ and $2\vec{p}-\vec{q}+2\vec{r}$. If the position vectors of the orthocenter and the circumcenter of the triangle are $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ and $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ respectively, then $\alpha + 2\beta + 5\gamma$ is equal to :

माना कि एक त्रिभुज के तीन शीर्षों के स्थिति सदिश $4\vec{p}+\vec{q}-3\vec{r}, -5\vec{p}+\vec{q}+2\vec{r}$ और $2\vec{p}-\vec{q}+2\vec{r}$ हैं। यदि त्रिभुज के लम्बकेन्द्र और परिकेन्द्र के स्थिति सदिश क्रमशः $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ और $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ हैं, तो $\alpha + 2\beta + 5\gamma$ बराबर है :

(1) 6 (2) 3 (3) 1 (4) 4

Ans. B

Sol.



$$\Rightarrow \frac{2(\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r})+\left(\frac{\vec{p}+\vec{q}+\vec{r}}{4}\right)}{3}=\frac{\vec{p}+\vec{q}+\vec{r}}{3}$$

On comparing : $2\alpha+\frac{1}{4}=1 \quad \Rightarrow \alpha=\frac{3}{8}$

$$\beta=\gamma=\frac{3}{8}$$

$\alpha + 2\beta + 5\gamma = 3$

22. Let $\vec{a}=3\hat{i}-\hat{j}+2\hat{k}$, $\vec{b}=\vec{a}\times\left(\hat{i}-2\hat{k}\right)$ and $\vec{c}=\vec{b}\times\hat{k}$. Then the projection of $\vec{c}-2\hat{j}$ on $\vec{a}$ is :

माना कि $\vec{a}=3\hat{i}-\hat{j}+2\hat{k}$, $\vec{b}=\vec{a}\times\left(\hat{i}-2\hat{k}\right)$ और $\vec{c}=\vec{b}\times\hat{k}$ है। तब $\vec{c}-2\hat{j}$ का $\vec{a}$ पर प्रक्षेप है :

(1) $\sqrt{14}$ (2) $2\sqrt{7}$ (3) $2\sqrt{14}$ (4) $3\sqrt{7}$

Ans. C

Sol. $\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & -1 & 2\\ 1 & 0 & -2\end{vmatrix}=2\hat{i}+8\hat{j}+\hat{k}$

$\vec{c}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 8 & 1\\ 0 & 0 & 1\end{vmatrix}=8\hat{i}-2\hat{j}$

Projection of $\left(\vec{c}-2\hat{j}\right)$ on $\vec{a}$

$\Rightarrow \dfrac{\left(8\hat{i}-4\hat{j}\right)\cdot\left(3\hat{i}-\hat{j}+2\hat{k}\right)}{\sqrt{9+1+4}}=2\sqrt{14}$

23. Let A, B, C be three points in xy-plane, whose position vector are given by $\sqrt{3}\hat{i}+\hat{j}$, $\hat{i}+\sqrt{3}\hat{j}$ and $a\hat{i}+(1-a)\hat{j}$ respectively with respect to the origin O. If the distance of the point C from the line bisecting the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\dfrac{9}{\sqrt{2}}$, then the sum of all the possible values of a is :

माना कि A, B, C, xy-तल में तीन बिन्दु हैं, जिनके स्थिति सदिश, मूल–बिन्दू O के सापेक्ष, क्रमशः $\sqrt{3}\hat{i}+\hat{j}$, $\hat{i}+\sqrt{3}\hat{j}$ और $a\hat{i}+(1-a)\hat{j}$ हैं। यदि बिन्दु C की, सदिशों $\overrightarrow{OA}$ और $\overrightarrow{OB}$ के बीच के कोण को सम–द्विभाजित करने वाली रेखा से दूरी $\dfrac{9}{\sqrt{2}}$, तो a के सभी संभव मानों का योग है :

(1) 0 (2) 1 (3) 2 (4) $\dfrac{9}{2}$

Ans. B

Sol. $\begin{bmatrix}\vec{A}=\sqrt{3}\hat{i}+\hat{j}\\ \vec{B}=\hat{i}+\sqrt{3}\hat{j}\\ \vec{C}=a\hat{i}+(1-a)\hat{j}\end{bmatrix}$

Angle bisector of $\overrightarrow{OA}$ and $\overrightarrow{OB}$ $\Rightarrow \widehat{OA}+\widehat{OB}$

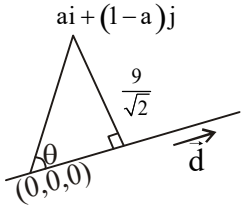
$\Rightarrow\left(\dfrac{\sqrt{3}\hat{i}+\hat{j}}{2}\right)+\dfrac{\left(\hat{i}+\sqrt{3}\hat{j}\right)}{2}$

$\Rightarrow \dfrac{\left(\sqrt{3}+1\right)\hat{i}+\left(\sqrt{3}+1\right)\hat{j}}{2}$

$\Rightarrow \dfrac{\left(\sqrt{3}+1\right)}{2}\left(\hat{i}+\hat{j}\right) \Rightarrow$ Direction vector angle bisector

$\Rightarrow \vec{r} \Rightarrow \vec{O}+\lambda\left(\vec{d}\right)$

$\vec{r} \Rightarrow \vec{O}+\lambda\left(\hat{i}+\hat{j}\right)$



$\left|\dfrac{\left(a\hat{i}+(1-a)\hat{j}\right)\times d}{\sqrt{a^2+(1-a)^2}\,|\vec{d}|}\right|\sqrt{a^2+(1-a)^2} \Rightarrow \left(\dfrac{9}{\sqrt{2}}\right)$

$\left|\dfrac{\left(a\hat{i}+(1-a)\hat{j}\right)\times\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}\right|=\dfrac{9}{\sqrt{2}}$

$\left|(a-1)\hat{k}+a\hat{k}\right|=9$

$|2a-1|=9$

$2a-1=9$ $\qquad$ $(2a-1)=-9$

$a=5$ $\qquad$ $2a=-8$

$a=-4$

Two possible values $\Rightarrow 5$ & -4

$\Rightarrow 5+(-4)=1$

24. If the components of $\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ along and perpendicular to $\vec{b}=3\hat{i}+\hat{j}-\hat{k}$ respectively, are $\dfrac{16}{11}\left(3\hat{i}+\hat{j}-\hat{k}\right)$ and $\dfrac{1}{11}\left(-4\hat{i}-5\hat{j}-17\hat{k}\right)$, then $\alpha^2+\beta^2+\gamma^2$ is equal to :

यदि $\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ के $\vec{b}=3\hat{i}+\hat{j}-\hat{k}$ के अनुदिश और लम्बवत् घटक क्रमशः $\frac{16}{11}\left(3\hat{i}+\hat{j}-\hat{k}\right)$ और $\frac{1}{11}\left(-4\hat{i}-5\hat{j}-17\hat{k}\right)$ हैं, तो $\alpha^2+\beta^2+\gamma^2$ बराबर है :

(1) 16 (2) 18 (3) 26 (4) 23

Ans. C

Sol. $\begin{bmatrix}\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}\\ \vec{b}=3\hat{i}+\hat{j}-\hat{k}\end{bmatrix}$

$\vec{a}=\vec{a}_b+\vec{a}_{b\perp}$

$\vec{a}_b \Rightarrow \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)\cdot\hat{b}$

$\Rightarrow\left(\vec{a}\cdot\vec{b}\right)=16$

$\Rightarrow\left(\vec{a}\cdot\vec{b}\right)\cdot\left(\frac{\vec{b}}{|\vec{b}|^2}\right)\Rightarrow\frac{(16)\left(3\hat{i}+\hat{j}-\hat{k}\right)}{\left(3^2+1^2+(-1)^2\right)}$

$\vec{a}_b=\frac{\left(\vec{a}\cdot\vec{b}\right)}{|\vec{b}|}\left(\frac{\vec{b}}{|\vec{b}|}\right)\Rightarrow 16\cdot\frac{\left(3\hat{i}+\hat{j}-\hat{k}\right)}{11}$

$3\alpha+\beta-\gamma=16$

$\vec{a}_{b\perp}=\vec{a}-\vec{a}_b$

$\Rightarrow\left(\alpha-\frac{48}{11}\right)\hat{i}+\left(\beta-\frac{16}{11}\right)\hat{j}+\left(\gamma+\frac{16}{11}\right)\hat{k}$

$=\left(\frac{-4}{11}\right)\hat{i}-\frac{5}{11}\hat{j}-\frac{17}{11}\hat{k}$

$\alpha-\frac{48}{11}=\left(-\frac{4}{11}\right)$

$\alpha=4$

$\beta-\frac{16}{11}=\left(-\frac{5}{11}\right)$

$\beta=1$

$\gamma+\frac{16}{11}=-\frac{17}{11}$

$\gamma=-3$

25. Let $\vec{a}=2\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}-5\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}\times\vec{c}=\vec{c}\times\vec{b}$ and $(\vec{a}+\vec{c}).(\vec{b}+\vec{c})=168$. Then the maximum value of $|\vec{c}|^2$ is :

माना कि $\vec{a}=2\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}-5\hat{j}+\hat{k}$ है और $\vec{c}$ एक सदिश इस प्रकार है कि $\vec{a}\times\vec{c}=\vec{c}\times\vec{b}$ और $(\vec{a}+\vec{c}).(\vec{b}+\vec{c})=168$ है। तब $|\vec{c}|^2$ का अधिकतम मान है :

(1) 308 (2) 154 (3) 77 (4) 462

Ans. A

Sol. $\vec{a}\times\vec{c}+\vec{b}\times\vec{c}=0$

$(\vec{a}+\vec{b})\times\vec{c}=0$

$\vec{c}=\lambda(\vec{a}+\vec{b})$

$\vec{c}=\lambda(5\hat{i}-6\hat{j}+4\hat{k})$

$\vec{c}=5\lambda\hat{i}-6\lambda\hat{j}+4\lambda\hat{k}$

$((2+5\lambda)\hat{i}-(1+6\lambda)\hat{j}+(4\lambda+3)\hat{k})\cdot((3+5\lambda)\hat{i}-(6\lambda+5)\hat{j}+(1+4\lambda)\hat{k})=168$

$(2+5\lambda)(3+5\lambda)+(1+6\lambda)(6\lambda+5)+(4\lambda+3)(1+4\lambda)=168$

$\lambda^2+\lambda-2=0$

$(\lambda+2)(\lambda-1)=0$

$\lambda=-2$ or $\lambda=1$

$|\vec{c}|=\sqrt{100+144+64}$

$|\vec{c}|^2=244+64=308$

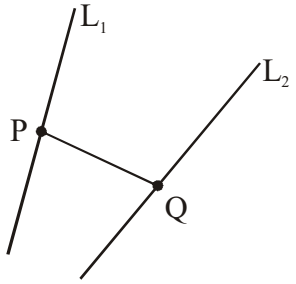
26. Let $L_1:\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $L_2:\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2 :

माना कि $L_1:\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ और $L_2:\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ दो रेखाएँ हैं। तब निम्नलिखित में से कौनसा बिन्दु L_1 और L_2 के बीच लघुतम दूरी की रेखा पर स्थित है ?

(1) $\left(-\frac{5}{3},-7,1\right)$ (2) $\left(\frac{8}{3},-1,\frac{1}{3}\right)$ (3) $\left(2,3,\frac{1}{3}\right)$ (4) $\left(\frac{14}{3},-3,\frac{22}{3}\right)$

Ans. D

Sol.



$P(2\lambda + 1, 3\lambda + 2, 4\lambda + 3)$ on L_1
$Q(3\mu + 2, 4\mu + 4, 5\mu + 5)$ on L_2
Dr's of PQ = $3\mu - 2\lambda + 1, 4\mu - 3\lambda + 2, 5\mu - 4\lambda + 2$
$PQ \perp L_1$
$\Rightarrow (3\mu - 2\lambda + 1)2 + (4\mu - 3\lambda + 2)3 + (5\mu - 4\lambda + 2)4 = 0$
$38\mu - 29\lambda + 16 = 0$ ...(1)
$PQ \perp L_2$
$\Rightarrow (3\mu - 2\lambda + 1)3 + (4\mu - 3\lambda + 2)4 + (5\mu - 4\lambda + 2)5 = 0$
$50\mu - 38\lambda + 21 = 0$ ...(2)

By (1) and (2)

$$\lambda = \frac{1}{3}; \mu = \frac{-1}{6}$$

$$\therefore P\left(\frac{5}{3}, 3, \frac{13}{3}\right) \text{ \& } Q\left(\frac{3}{2}, \frac{10}{3}, \frac{25}{6}\right)$$

Line PQ

$$\frac{x - \frac{5}{3}}{\frac{1}{6}} \qquad \frac{y - 3}{\frac{-1}{3}} \qquad \frac{z - \frac{13}{3}}{\frac{1}{6}}$$

$$\frac{x - \frac{5}{3}}{1} = \frac{y - 3}{-2} = \frac{z - \frac{13}{3}}{1}$$

Point $\left(\frac{14}{3}, -3, \frac{22}{3}\right)$
lies on the line PQ.

27. Let in a ΔABC, the length of the side AC be 6, the vertex B be (1, 2, 3) and the vertices A, C lie on the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Then the area (in sq. units) of ΔABC is :

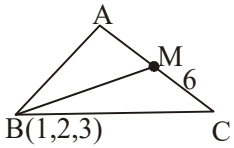
माना कि एक ΔABC में भुजा AC की लम्बाई 6 है, शीर्ष B(1, 2, 3) है और शीर्ष A, C रेखा $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ पर स्थित हैं।

तब ΔABC का क्षेत्रफल (वर्ग इकाई में) है :

(1) 17 (2) 21 (3) 56 (4) 42

Ans. B

Sol.



Let M $(3\lambda + 6, 2\lambda + 7, -2\lambda + 7)$

$\overrightarrow{BM} = (3\lambda+5)\hat{i} + (2\lambda+5)\hat{j} + (-2\lambda+4)\hat{k}$

$\overrightarrow{AC}\cdot\overrightarrow{BM} = 0 = 3(3\lambda + 5) + 2(2\lambda + 5) - 2(-2\lambda + 4)$

$\overrightarrow{BM} = 2\hat{i} + 3\hat{j} + 6\hat{k}$

$\left|\overrightarrow{BM}\right| = 7$

$\text{Area} = \frac{1}{2} \times 6 \times 7 = 21$

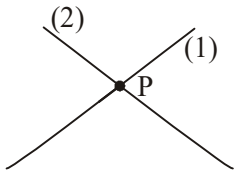
28. Let the line passing through the points (–1, 2, 1) and parallel to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ intersect the line $\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ at the point P. Then the distance of P from the point Q(4, –5, 1) is :

माना कि बिन्दु (–1, 2, 1) से होकर गुजरती और रेखा $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ के समान्तर रेखा, रेखा $\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ को बिन्दु P पर प्रतिच्छेद करती है। तब बिन्दु Q(4, –5, 1) से P की दूरी है :

(1) $5\sqrt{5}$ (2) 10 (3) 5 (4) $5\sqrt{6}$

Ans. A

Sol.



Equation of line through point (–1, 2, 1) is $\rightarrow$

$\Rightarrow \frac{x+1}{2} = \frac{y-2}{3} = \frac{z-1}{4} = \lambda$

So, $\begin{cases} x = 2\lambda - 1 \\ y = 3\lambda + 2 \\ z = 4\lambda + 1 \end{cases}$

By (1) $\rightarrow \frac{x+2}{3}=\frac{y-3}{2}=\frac{z-4}{1}=\mu$ (Let)

So, $\begin{bmatrix} x=3\mu-2 \\ y=2\mu+3 \\ z=\mu+4 \end{bmatrix}$

For intersection point 'P'

$x = 2\lambda - 1 = 3\mu - 2$

$y = 3\lambda + 2 = 2\mu + 3 \quad \begin{bmatrix} \lambda=1 \\ \mu=1 \end{bmatrix}$

$z = 4\lambda + 1 = \mu + 4$

So, point P(x, y, z) = (1, 5, 5)

W& Q(4, –5, 1)

$\therefore PQ = \sqrt{9+100+16}$

$= \sqrt{125} = 5\sqrt{5}$

29. Let A (x, y, z) be a point in xy-plane, which is equidistant from three points (0, 3, 2), (2, 0, 3) and (0, 0, 1). Let B = (1, 4, –1) and C = (2, 0, –2). Then among the statements

(S1) : ΔABC is an isosceles right angled triangle, and

(S2) : the area of ΔABC is $\frac{9\sqrt{2}}{2}$:

(1) both are true (2) both are false (3) only (S2) is true (4) only (S1) is true

माना कि xy-तल में एक बिन्दु A (x, y, z) है जो कि तीन बिन्दुओं (0, 3, 2), (2, 0, 3) और (0, 0, 1) से समदूरस्थ है। माना कि B = (1, 4, –1) और C = (2, 0, –2) है। तब कथनों

(S1) : ΔABC एक समद्विबाहु समकोण त्रिभुज है, और

(S2) : ΔABC का क्षेत्रफल $\frac{9\sqrt{2}}{2}$ है

में से

(1) दोनों सत्य हैं (2) दोनों असत्य हैं (3) केवल (S2) सत्य है (4) केवल (S1) सत्य है

Ans. D

Sol. A(x, y, z) Let P(0,3,2), Q(2,0,3), R(0,0,1)

AP = AQ = AR

$x^2 + (y-3)^2 + (z-2)^2 = (x-2)^2 + y^2 + (z-3)^2 = x^2 + y^2 + (z-1)^2$

In xy-plane z = 0

So, $x^2 - 4x + 4 + y^2 + 9 = x^2 + y^2 + 1$

$x = 3$

$9 + y^2 - 6y + 9 + 4 = x^2 + y^2 + 1$

So, A(3,2,0) also B(1,4,–1) & C(2,0,–2)

Now, $AB = \sqrt{4+4+1} = 3$

$AC = \sqrt{1+4+4} = 3$

$BC = \sqrt{1+16+1} = \sqrt{18}$

$AB = AC$

isosceles Δ & $AB^2 + AC^2 = BC^2$

Area of $\Delta ABC = \frac{1}{2} \times$ base.height

$\frac{1}{2} \times 3 \times 3 = \frac{9}{2}$

So, only S_1 is true

30. Let $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$ and $L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$ be two lines. Let L_3 be a line passing through the point (α, β, γ) and be perpendicular to both L_1 and L_2. If L_3 intersects L_1, then $|5\alpha - 11\beta - 8\gamma|$ is equals :

माना कि $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$ और $L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$ दो रेखाऐं हैं। माना L_3, बिन्दु (α, β, γ) से गुजरने वाली और L_1 तथा L_2 दोनों के लम्बवत् एक रेखा है। यदि L_3, L_1 को प्रतिच्छेद करती है, तो $|5\alpha - 11\beta - 8\gamma|$ बराबर है :

(1) 16 (2) 25 (3) 20 (4) 18

Ans. B

Sol. $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$

$L_1 : \vec{r} = (1,2,1) + \lambda(1,-1,2)$

$L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$

$L_2 : \vec{r} = (-1,2,0) + \mu(-1,2,1)$

$$\vec{P} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ -1 & 2 & 1 \end{vmatrix}$$

$= \hat{i}(-5) - \hat{j}(3) + \hat{k}(1)$

$= -5\hat{i} - 3\hat{j} + \hat{k}$

$P(\alpha, \beta, \gamma)$ $\quad Q(1 + \lambda, 2 - \lambda, 1 + 2\lambda)$

$\overrightarrow{PQ} = (1+\lambda-\alpha, 2-\lambda-\beta, 1+2\lambda-\gamma)$

$1 + \lambda - \alpha = -5\mu$ $\quad 2 - \lambda - \beta = -3\mu$ $\quad 1 + 2\lambda - \gamma = \mu$

$1 + \lambda + 5\mu = \alpha$ $\quad 2 - \lambda + 3\mu = \beta$ $\quad \gamma = 1 + 2\lambda - \mu$

$|5\alpha - 11\beta - 8\gamma| = 25$

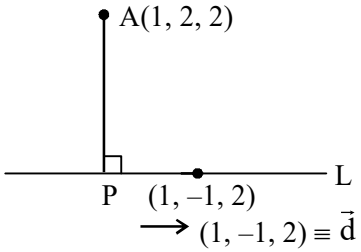
31. Let P be the foot of the perpendicular from point (1, 2, 2) on the line $L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the line $\vec{r} = \left(-\hat{i} + \hat{j} - 2\hat{k}\right) + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right), \lambda \in R$, intersect the line L, at Q. Then $2(PQ)^2$ is equal to :

माना कि P, बिन्दु (1, 2, 2) से रेखा $L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$ पर लम्ब का पाद है। माना कि रेखा $\vec{r} = \left(-\hat{i} + \hat{j} - 2\hat{k}\right) + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right), \lambda \in R$, रेखा L को Q पर काटती है। तब $2(PQ)^2$ बराबर है :

(1) 19 $\quad$ (2) 27 $\quad$ (3) 25 $\quad$ (4) 29

Ans. B

Sol.



$L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2} = \mu$

$= \mu P(\mu + 1, -\mu - 1, 2\mu + 2)$

$\overrightarrow{AP} \cdot \vec{d} = 0 \quad \Rightarrow (\mu, -\mu - 3, 2\mu) \cdot (1, -1, 2) = 0$

$\Rightarrow \mu + \mu + 3 + 4\mu = 0 \quad \Rightarrow \mu = -\frac{1}{2}$

$\therefore P\left(\frac{-1}{2}+1, +\frac{1}{2}-1, 2\left(\frac{-1}{2}\right)+2\right)$

$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right)$

Now general pt. on L_2 is $Q(-1 + \lambda, 1 - \lambda, -2 + \lambda)$

Equate it with general pt of L

$\mu + 1 = -1 + \lambda$ | $-\mu - 1 = 1 - \lambda$ | $2\mu + 2 = -2 + \lambda$

$\mu = \lambda - 2$ | $\mu = \lambda - 2$ | $\downarrow$

$2(\lambda - 2) + 2 = -2 + \lambda$

$2\lambda - 4 + 2 = -2 + \lambda$

$\therefore \mu = -2, \lambda = 0$

$\therefore Q \equiv -2, \lambda = 0$

$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right)$ and $Q(-1,1,-2)$

$$PQ = \sqrt{\left(\frac{1}{2}+1\right)^2 + \left(\frac{-1}{2}-1\right)^2 + (1+2)^2}$$

$$= \sqrt{\frac{9}{4}+\frac{9}{4}+9} = \sqrt{\frac{54}{4}}$$

$$\therefore 2(PQ)^2 = 2\left(\frac{54}{4}\right) = 27$$

32. Let a unit vector which makes an angle of 60º with $2\hat{i}+2\hat{j}-\hat{k}$ and an angle of 45º with $\hat{i}-\hat{k}$ be $\vec{C}$. Then $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ is :

माना सदिश $2\hat{i}+2\hat{j}-\hat{k}$ से 60º का कोण तथा सदिश $\hat{i}-\hat{k}$ से 45° का कोण बनाने वाला एक मात्रक सदिश $\vec{C}$ है। तो $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ है :

(1) $\left(\frac{1}{\sqrt{3}}+\frac{1}{2}\right)\hat{i}+\left(\frac{1}{\sqrt{3}}-\frac{1}{3\sqrt{2}}\right)\hat{j}+\left(\frac{1}{\sqrt{3}}+\frac{\sqrt{2}}{3}\right)\hat{k}$

(2) $\frac{\sqrt{2}}{3}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{1}{2}\hat{k}$

(3) $\frac{\sqrt{2}}{3}\hat{i}-\frac{1}{2}\hat{k}$

(4) $-\frac{\sqrt{2}}{3}\hat{i}+\frac{\sqrt{2}}{3}\hat{j}+\left(\frac{1}{2}+\frac{2\sqrt{2}}{3}\right)\hat{k}$

Ans. C

Sol. Let $\vec{C} = a\hat{i}+b\hat{j}+c\hat{k}$

$$\left(a\hat{i}+b\hat{j}+c\hat{k}\right)\cdot\left(2\hat{i}+2\hat{j}-\hat{k}\right) = 1\times 3\times\frac{1}{2}$$

$$2a+2b-c = \frac{3}{2} \qquad \text{_____(1)}$$

$$\left(a\hat{i}+b\hat{j}+c\hat{k}\right)\cdot\left(\hat{i}-\hat{k}\right) = 1\times\sqrt{2}\times\frac{1}{\sqrt{2}}$$

$a - c = 1$ _____(2)

$a^2 + b^2 + c^2 = 1$ _____(3)

Solving (1), (2) and (3)

$$a + 2b = \frac{1}{2}$$

$a^2 + b^2 + (a-1)^2 = 1$

$2a^2 - 2a + b^2 = 0$

$$2a^2 - 2a + \left(\frac{2a-1}{4}\right)^2 = 0$$

$32a^2 - 32a + 4a^2 - 4a + 1 = 0$

$36a^2 - 36a + 1 = 0$

$$a = \frac{36 \pm \sqrt{(36)^2 - 4(36)}}{2 \times 36}$$

$$= \frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$b = \frac{1-2a}{4} \Rightarrow b = \frac{1 \pm \frac{2\sqrt{2}}{3} - 1}{4}$$

$$= -\pm \frac{1}{3\sqrt{2}}$$

$$C = -\frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$C + \left(\frac{-1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$$

$$= \frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$$

33. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$, $x \in R$. If $\vec{d}$ is the unit vector in the direction of $\vec{b} + \vec{c}$ such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to :

माना $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ तथा $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$, $x \in R$ है। यदि $\vec{b} + \vec{c}$ की दिशा में एक इकाई सदिश $\vec{d}$ इस प्रकार है कि $\vec{a} \cdot \vec{d} = 1$, तो $(\vec{a} \times \vec{b}) \cdot \vec{c}$ बराबर है :

(1) 11 (2) 3 (3) 6 (4) 9

Ans. A

Sol. $\vec{d} = \lambda\left((2+x)\hat{i} + 6\hat{j} - 2\hat{k}\right)$

$\vec{a}\cdot\vec{d} = 1 \Rightarrow \lambda(x+6) = 1$ _____(1)

and $|\vec{d}| = 1 \Rightarrow \lambda^2\left((x+2)^2 + 36 + 4\right) = 1$ _____(2)

(1) and (2) : x = 1

$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{a}\times\vec{b} = -9\hat{i} + 7\hat{j} + 2\hat{k}$

$\left(\vec{a}\times\vec{b}\right)\cdot\vec{c} = -9 + 14 + 6 = 11$

34. Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b}\times\vec{c}$. If $\alpha \in \left[0, \frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to :

तीन सदिशों $\vec{a}, \vec{b}, \vec{c}$ का विचार कीजिए। माना $|\vec{a}| = 2, |\vec{b}| = 3$ तथा $\vec{a} = \vec{b}\times\vec{c}$ हैं। यदि सदिशों $\vec{b}$ तथा $\vec{c}$ के बीच का कोण $\alpha \in \left[0, \frac{\pi}{3}\right]$ है, तो $27|\vec{c} - \vec{a}|^2$ का न्यूनतम मान है :

(1) 105 (2) 124 (3) 121 (4) 110

Ans. B

Sol. $\vec{a} = \vec{b}\times\vec{c}$

$|\vec{a}| = 2, |\vec{b}| = 3$

$\vec{a}.\vec{b} = 0$ and $\vec{a}.\vec{c} = 0$

$|\vec{c} - \vec{a}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c}\cdot\vec{a}$

$= 4 + |\vec{c}|^2$

$|\vec{a}| = |\vec{b}\times\vec{c}| = |\vec{b}|\sin\alpha\,|\vec{c}|$

$\Rightarrow \sin\alpha\,|\vec{c}| = \frac{2}{3}$

$\Rightarrow \sin^2\alpha = \dfrac{4}{9|\vec{c}|^2}$

$\Rightarrow |\vec{c}|^2 = \dfrac{4}{9\sin^2\alpha}$

$\Rightarrow |\vec{c}-\vec{a}|^2 = 4 + \dfrac{4}{9\sin^2\alpha}$

For $|\vec{c}-\vec{a}|^2$ to be minimum for $\alpha \in \left[0, \dfrac{\pi}{3}\right]$

$\sin\alpha = \dfrac{\sqrt{3}}{2}$

$27|\vec{c}-\vec{a}|^2 = 27\left[4 + \dfrac{4.4}{9.3}\right]$

$= 27\left[\dfrac{124}{27}\right] = 124$

35. Let $\vec{a} = 6\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a is vector such that $|\vec{c}| \geq 6, \vec{a}\cdot\vec{c} = 6|\vec{c}|, |\vec{c}-\vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|(\vec{a}\times\vec{b})\times\vec{c}\right|$ is equal to :

माना $\vec{a} = 6\hat{i} + \hat{j} - \hat{k}$ तथा $\vec{b} = \hat{i} + \hat{j}$ है। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $|\vec{c}| \geq 6, \vec{a}\cdot\vec{c} = 6|\vec{c}|, |\vec{c}-\vec{a}| = 2\sqrt{2}$ तथा $\vec{a}\times\vec{b}$ और $\vec{c}$ के मध्य का कोण 60° है, तो $\left|(\vec{a}\times\vec{b})\times\vec{c}\right|$ बराबर है :

(1) $\dfrac{9}{2}\left(6-\sqrt{6}\right)$ (2) $\dfrac{9}{2}\left(6+\sqrt{6}\right)$ (3) $\dfrac{3}{2}\sqrt{3}$ (4) $\dfrac{3}{2}\sqrt{6}$

Ans. B

Sol. $|(\vec{a}\times\vec{b})\times\vec{c}| = |\vec{a}\times\vec{b}||\vec{c}|\dfrac{\sqrt{3}}{2}$

$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i} - \hat{j} + 5\hat{k}$

$|\vec{a}\times\vec{b}| = \sqrt{27}$

$\therefore\ |\vec{c}-\vec{a}| = 2\sqrt{2}$

$|\vec{c}|^2 + |\vec{a}| - 2\vec{c}.\vec{a} = 8$

$|\vec{c}|^2 + |\vec{a}| - 12|\vec{c}| = 8$

$|\vec{c}|^2 - 12|\vec{c}| + 30 = 0$

$|\vec{c}| = 6 + \sqrt{6}$

$\therefore \ |\vec{c}| > 6$

$$\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right| = \sqrt{27}\times\left(6+\sqrt{6}\right)\times\frac{\sqrt{3}}{2} = \frac{9}{2}\left(6+\sqrt{6}\right)$$

36. Let P(x, y, z) be a point in the first octant, whose projection in the xy-plane is the point Q. Let OP = γ; the angle between OQ and the positive x-axis is θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is :

माना P(x, y, z) प्रथम अष्टांशक में एक बिन्दु है जिसका xy-तल में प्रक्षेप बिन्दु Q है। माना OP = γ है तथा OQ और धनात्मक x-axis कोण θ है एवं OP तथा धनात्मक z-अक्ष के बीच कोण ϕ है, जहाँ O मूल बिन्दु है, तब P की x-अक्ष से दूरी होगी :

(1) $\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$ (2) $\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

(3) $\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$ (4) $\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$

Ans. C

Sol. $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\overrightarrow{OQ} = x\hat{i} + y\hat{j}$

$|OP| = \gamma = \sqrt{x^2+y^2+z^2}$

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}} \Rightarrow \cos^2\theta = \frac{x^2}{\gamma^2-z^2} = \frac{x^2}{\gamma^2-\gamma^2\cos^2\phi}$$

$$\cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}} = \frac{z}{\gamma}$$

Distance of P from x-axis = $\sqrt{y^2+z^2}$

$d = \sqrt{\gamma^2 - x^2}$

$\Rightarrow x^2 = \gamma^2 \sin^2\phi \cos^2\theta$

$\Rightarrow d = \sqrt{\gamma^2 - \gamma^2\sin^2\phi\cos^2\theta}$

$= \gamma\sqrt{1-\sin^2\phi\cos^2\theta}$

37. Let $\vec{a} = \hat{i}+2\hat{j}+3\hat{k}$, $\vec{b} = 2\hat{i}+3\hat{j}-5\hat{k}$ and $\vec{c} = 3\hat{i}-\hat{j}+\lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b}+\vec{c}$. If $\vec{r}\cdot\vec{a} = 3$, then 3λ is equal to :

माना तीन सदिश $\vec{a} = \hat{i}+2\hat{j}+3\hat{k}$, $\vec{b} = 2\hat{i}+3\hat{j}-5\hat{k}$ तथा $\vec{c} = 3\hat{i}-\hat{j}+\lambda\hat{k}$ हैं। माना $\vec{b}+\vec{c}$ के अनुदिश एक मात्रक सदिश $\vec{r}$ है। यदि $\vec{r}\cdot\vec{a} = 3$ है, तो 3λ बराबर है :

(1) 30 (2) 21 (3) 25 (4) 27

Ans. C

Sol. $\vec{r} = k\left(\vec{b}+\vec{c}\right)$

$\vec{r}\cdot\vec{a} = 3$

$\vec{r}\cdot\vec{a} = k\left(\vec{b}\cdot\vec{a}+\vec{c}\cdot\vec{a}\right)$

$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$

$3 = k(-3 + 3\lambda)$ ...(1)

$\vec{r} = k\left(5\hat{i}-2\hat{j}-(5-\lambda)\hat{k}\right)$

$|\vec{r}| = k\sqrt{25+4+25+\lambda^2-10\lambda}$

$k = \dfrac{3}{-6+3\lambda} = \dfrac{1}{-2+\lambda}$ put in (1)

$4 + \lambda^2 - 4\lambda = (54 + \lambda^2 - 10\lambda)$

$6\lambda = 50$

$3\lambda = 25$

38. Let three vectors $\vec{a} = \alpha\hat{i}+4\hat{j}+2\hat{k}, \vec{b} = 5\hat{i}+3\hat{j}+4\hat{k}, \vec{c} = x\hat{i}+y\hat{i}+z\hat{k}$ form a triangle such that $\vec{c} = \vec{a}-\vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to :

माना तीन सदिश $\vec{a} = \alpha\hat{i}+4\hat{j}+2\hat{k}, \vec{b} = 5\hat{i}+3\hat{j}+4\hat{k}, \vec{c} = x\hat{i}+y\hat{i}+z\hat{k}$ एक त्रिभुज बनाते है, जो इस प्रकार है कि $\vec{c} = \vec{a}-\vec{b}$ है तथा इस त्रिभुज का क्षेत्रफल $5\sqrt{6}$ है। यदि α एक धनात्मक वास्तविक संख्या है, तो $|\vec{c}|^2$ बराबर है :

(1) 14 (2) 10 (3) 12 (4) 16

Ans. A

Sol. $c = (\alpha-5)\hat{i}+\hat{j}-2\hat{k}$

$$\text{Area of } \Delta ABC = \left|\frac{1}{2}\left(\vec{a}\times\vec{b}\right)\right| = 5\sqrt{6} \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 4 & 2 \\ 5 & 3 & 4 \end{vmatrix} = 10\sqrt{6}$$

$$\left|10\hat{i}-(4\alpha-10)\hat{j}+(3\alpha-20)\hat{k}\right| = 10\sqrt{6}$$

$$\Rightarrow \sqrt{100+(10-4\alpha)^2+(3\alpha-20)^2} = 10\sqrt{6}$$

$$\Rightarrow 25\alpha^2 - 200\alpha + 600 = 600$$

$$\Rightarrow \alpha = 8 \qquad (\because a \in R^+)$$

$$\vec{c} = 3\hat{i}+\hat{j}-2\hat{k}$$

$$\Rightarrow |\vec{c}|^2 = 9+1+4 = 14$$

39. Between the following two statements :

Statement I : Let $\vec{a} = \hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b} = 2\hat{i}+\hat{j}-\hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a}\times\vec{r} = \vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{r} = 0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC, $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$.

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

निम्नलिखित दो कथनों :

कथन I : माना $\vec{a} = \hat{i}+2\hat{j}-3\hat{k}$ और $\vec{b} = 2\hat{i}+\hat{j}-\hat{k}$ हैं। तो $\vec{a}\times\vec{r} = \vec{a}\times\vec{b}$ तथा $\vec{a}\cdot\vec{r} = 0$ को संतुष्ट करने वाले सदिश $\vec{r}$ का परिमाण $\sqrt{10}$ है।

कथन II : एक त्रिभुज ABC में $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$ है।

(1) कथन I सही है परन्तु कथन II गलत है।

(2) कथन I गलत है पररन्तु II सही है।

(3) कथन I तथा कथन II दोनों सही है।

(4) कथन I तथा कथन II गलत है।

Ans. B

Sol. $\forall$ two vectors $\vec{c}$ and $\vec{d}$

$$\left|\vec{c}\times\vec{d}\right|^2 = \left|\vec{c}\right|^2\left|\vec{d}\right|^2 - \left(\vec{c}\cdot\vec{d}\right)^2$$

replacing $\vec{c}=\vec{a}$ and $\vec{d}=\vec{r}$

$$\Rightarrow \left|\vec{a}\times\vec{r}\right| = \left|\vec{a}\right|^2\left|\vec{r}\right|^2 - \left(\vec{a}\cdot\vec{r}\right)^2$$

$$\Rightarrow \quad \left|\vec{a}\times\vec{b}\right| = \left|\vec{a}\right|^2\left|\vec{r}\right|^2$$

$$\left(\because \vec{a}\times\vec{r} = \vec{a}\times\vec{b} \text{ and } \vec{a}\cdot\vec{r}=0\right)$$

$$\Rightarrow 35 = 14\left|\vec{r}\right|^2$$

$$\Rightarrow \left|\vec{r}\right| = \sqrt{\frac{35}{14}} = \sqrt{\frac{5}{2}} \neq \sqrt{10}$$

$\therefore$ Statement {I} is incorrect Statement II is correct

$$\left(\text{i.e., } \cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}\right)$$

Proof : $\because \left(\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}\right)\geq 0$ _____(1)

and $\left|\overrightarrow{OA}\right|^2 = \left|\overrightarrow{OB}\right|^2 = \left|\overrightarrow{OC}\right|^2 = R^2$ _____(2)

Now, using (1), we get $\left|\overrightarrow{OA}\right|^2 + \left|\overrightarrow{OB}\right|^2 + \left|\overrightarrow{OC}\right|^2 + 2\left(\overrightarrow{OA}\cdot\overrightarrow{OB} + \overrightarrow{OB}\cdot\overrightarrow{OC} + \overrightarrow{OC}\cdot\overrightarrow{OA}\right)\geq 0$

$\Rightarrow 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C) \geq 0$

$$\Rightarrow \cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$$

40. Let $\vec{a} = 2\hat{i}+\alpha\hat{j}+\hat{k}$, $\vec{b} = -\hat{i}+\hat{k}$, $\vec{c} = \beta\hat{j}-\hat{k}$, where α and β are integers and $\alpha\beta = -6$. Let the values of the ordered pair (α, β), for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1, β_1) and (α_2, β_2). Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to:

माना $\vec{a} = 2\hat{i}+\alpha\hat{j}+\hat{k}$, $\vec{b} = -\hat{i}+\hat{k}$, $\vec{c} = \beta\hat{j}-\hat{k}$ हैं, जहाँ α और β पूर्णांक हैं और $\alpha\beta = -6$ है। माना क्रमित युग्म (α, β) के मान जिनके लिए विकर्णों $\vec{a}+\vec{b}$ और $\vec{b}+\vec{c}$ के समांतर चतुर्भुज का क्षेत्रफल $\frac{\sqrt{21}}{2}$ है, (α_1, β_1) और (α_2, β_2) हैं। तो $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$

बराबर है :

(1) 21 (2) 17 (3) 19 (4) 24

Ans. C

Sol. Area of parallelogram $= \frac{1}{2}\left|\vec{d}_1 \times \vec{d}_2\right.$

$$A = \frac{1}{2}\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right|$$

$$= \frac{\sqrt{21}}{2}$$

so, $\vec{a}+\vec{b} = \hat{i}+\alpha\hat{j}+2\hat{k}$

$\vec{b}+\vec{c} = -\hat{i}+\beta\hat{j}$

$$\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha & 2 \\ -1 & \beta & 0 \end{vmatrix}$$

$= \hat{i}(-2\beta) - \hat{j}(2) + \hat{k}(\beta+\alpha)$

$\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right| = \sqrt{4\beta^2+4+(\alpha+\beta)^2} = \sqrt{21}$

$4\beta^2 + 4 + \alpha^2 + \beta^2 + 2\alpha\beta = 21$

$\alpha^2 + 5\beta^2 - 12 = 17$

$\alpha^2 + 5\beta^2 = 29$ and $\alpha\beta = -6$

and $\alpha\beta = -6$

and given $\alpha_i\beta$ are integers

so,

$\alpha = -3, \beta = 2$

or

$\alpha = 3, \beta = -2$

$(\alpha_1, \beta_1) = (-3, 2)$

$(\alpha_2, \beta_2) = (3, -2)$

$\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2 = 9 + 4 + 6 = 19$

41. Let the point, on the line passing through the points P(1, –2,3) and Q(5, –4, 7), farther from the origin and at a distance of 9 units from the point P, be (α, β, γ). Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

माना बिन्दुओं P(1, –2,3) तथा Q(5, –4, 7) से होकर जाने वाली रेखा पर, मूल बिन्दु से ज्यादा दूर तथा बिन्दु P से 9 इकाई दूरी पर, स्थित बिन्दु (α, β, γ) है, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है :

(1) 155 (2) 150 (3) 165 (4) 160

Ans. A

Sol. PQ line

$$\frac{x-1}{4} = \frac{y+2}{-2} = \frac{z-3}{4}$$

pt$(4t + 1, -2t - 2, 4t + 3)$

distance$^2 = 16t^2 + 4t^2 + 16t^2 = 81$

$$t = \pm\frac{3}{2}$$

pt$(7, -5, 9)$

$\alpha^2 + \beta^2 + \gamma^2 = 155$

42. Let the line L intersect the lines $x - 2 = -y = z - 1$, $2(x + 1) = 2(y - 1) = z + 1$ and be parallel to the line $\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}$. Then which of the following points lies on L?

माना रेखा L, रेखाओं $x - 2 = -y = z - 1$, $2(x + 1) = 2(y - 1) = z + 1$ को काटती है, तथा रेखा $\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}$ के समान्तर है। तो निम्न में से कौनसा बिन्दु L पर है?

(1) $\left(-\frac{1}{3}, -1, -1\right)$ (2) $\left(-\frac{1}{3}, -1, 1\right)$ (3) $\left(-\frac{1}{3}, 1, -1\right)$ (4) $\left(-\frac{1}{3}, 1, 1\right)$

Ans. C

Sol. $\frac{x-2}{1} = \frac{y}{-1} = \frac{z-1}{1}, \frac{x+1}{1/2} = \frac{y-1}{1/2} = \frac{z+1}{1}$

$(k + 2, -k, k + 1)$, $\left(\frac{\alpha}{2} - 1, \frac{\alpha}{2} + 1, \alpha - 1\right)$

$$\frac{\frac{\alpha}{2} - 1 - k - 2}{3} = \frac{\frac{\alpha}{2} + 1 + k}{1} \quad \frac{\alpha - 1 - k - 1}{2}$$

$$k = -\frac{4}{3}$$

$\therefore$ L passes through point $\left(\frac{2}{3},\frac{4}{3},-\frac{1}{3}\right)$.

$$L:\frac{x-\frac{2}{3}}{3}=\frac{y-\frac{4}{3}}{1}=\frac{z+\frac{1}{3}}{2}$$

$\left(-\frac{1}{3},1,-1\right)$ lies on L

43. Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=3\left(\hat{i}-\hat{j}+\hat{k}\right)$. Let $\vec{c}$ be the vector such that $\vec{a}\times\vec{c}=\vec{b}$ and $\vec{a}\cdot\vec{c}=3$. Then $\vec{a}.\left(\left(\vec{c}\times\vec{b}\right)-\vec{b}-\vec{c}\right)$ is equal to :

माना $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=3\left(\hat{i}-\hat{j}+\hat{k}\right)$ है। माना एक सदिश $\vec{c}$ के लिए $\vec{a}\times\vec{c}=\vec{b}$ तथा $\vec{a}\cdot\vec{c}=3$ हैं। तो $\vec{a}.\left(\left(\vec{c}\times\vec{b}\right)-\vec{b}-\vec{c}\right)$ बराबर है :

(1) 36 (2) 24 (3) 32 (4) 20

Ans. B

Sol. $(\vec{a}\times\vec{c}).\vec{b}=\vec{b}\cdot\vec{b}$

$\left[\vec{a}\,\vec{c}\,\vec{b}\right]=9(1+1+1)=27$

$\vec{a}\cdot\vec{b}=3(1-2+1)=0$

$\vec{a}\cdot\vec{c}=3$

$\vec{a}\cdot((\vec{c}\times\vec{b})-\vec{b}-\vec{c})=[\vec{a}\vec{c}\vec{b}]-\vec{a}\cdot\vec{b}-\vec{a}\cdot\vec{c}$

$= 27 - 0 - 3 = 24$

44. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$ respectively. Let ℓ_1,ℓ_2 and ℓ_3 be be the lengths of perpendiculars drawn from the ortho center of the triangle on the sides AB, BC and CA respectively, then $\ell_1^2+\ell_2^2+\ell_3^2$ equal :

माना एक त्रिभुज के शीर्षो A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+2\hat{k}$ तथा $2\hat{i}+\hat{j}+2\hat{k}$ हैं। माना इस त्रिभुज के लम्ब केन्द्र से भुजाओं AB, BC तथा CA पर डाले गए लम्बों की लम्बाईयाँ क्रमशः ℓ_1,ℓ_2 तथा ℓ_3 हैं, तो $\ell_1^2+\ell_2^2+\ell_3^2$ बराबर है :

(1) $\frac{1}{5}$ (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{1}{3}$

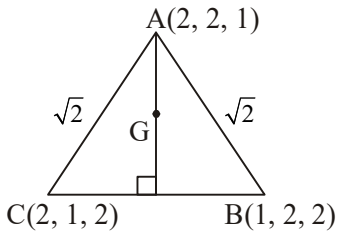
Ans. B

Sol. Equilateral Δ

$$\sqrt{2-\frac{1}{2}}=\sqrt{\frac{3}{2}}$$

$$\frac{1}{3}\times\sqrt{\frac{3}{2}}=\frac{1}{\sqrt{6}}$$

$$\ell_1^2+\ell_2^2+\ell_3^2=3\ell_1^2=3\times\frac{1}{6}=\frac{1}{2}$$



45. The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let *l* denotes the length of the angle bisector AD of ∠BAC where D is on the line segment BC, then $2l^2$ equals :

एक त्रिभुज के शीर्षो A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ तथा $-\hat{i}+\hat{j}+3\hat{k}$ हैं। माना कोण ∠BAC के समद्विभाजक AD की लम्बाई *l* है, जहाँ बिंदु D, रेखाखण्ड BC पर है, तो $2l^2$ बराबर है :

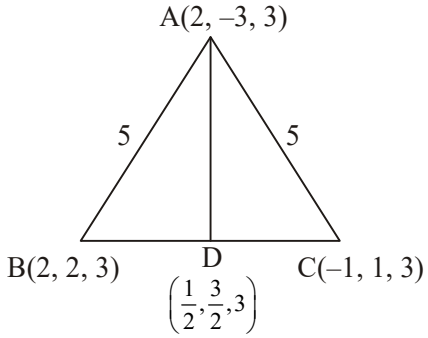
(1) 49 (2) 42 (3) 50 (4) 45

Ans. D

Sol. $\ell^2=\left(2-\frac{1}{2}\right)^2+\left(3+\frac{3}{2}\right)^2+0$

$$\ell^2=\frac{9}{4}+\frac{81}{4}$$

$$\ell^2=\frac{45}{2} \qquad 2\ell^2=45$$



46. Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of ∠AOB meets the line AB at C, then the length of OC is :

माना O मूलबिन्दु है तथा A और B के स्थिति सदिश क्रमशः $2\hat{i}+2\hat{j}+\hat{k}$ और $2\hat{i}+4\hat{j}+4\hat{k}$ हैं। यदि ∠AOB की अंतः समद्विभाजक रेखा AB को C पर मिलती है, तो OC की लंबाई है :

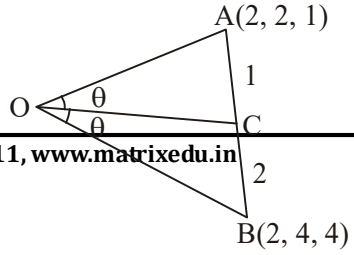
(1) $\frac{2}{3}\sqrt{34}$ (2) $\frac{3}{2}\sqrt{31}$ (3) $\frac{2}{3}\sqrt{31}$ (4) $\frac{3}{2}\sqrt{34}$

Ans. A

Sol. $\left|\overrightarrow{OA}\right|=\sqrt{4+4+1}=3$

$\left|\overrightarrow{OB}\right|=\sqrt{4+16+16}=6$

AC : CB = 3 : 6 = 1 : 2



$$c\left(\frac{4+2}{3},\frac{4+4}{3},\frac{2+4}{3}\right)$$

$$c\left(2,\frac{8}{3},2\right)$$

$$\left|\overrightarrow{OC}\right| = \sqrt{4+\frac{64}{9}+4} = \frac{\sqrt{136}}{3} = \frac{2}{3}\sqrt{34}$$

47. Let a unit vector $\vec{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ then $\left|\hat{u} - \vec{v}\right|^2$ is equal to :

माना एक मात्रक सदिश $\vec{u} = x\hat{i} + y\hat{j} + z\hat{k}$, सदिशों $\frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ तथा $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ से क्रमशः

$\frac{\pi}{2}$, $\frac{\pi}{3}$ तथा $\frac{2\pi}{3}$ के कोण बनाता है। यदि $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ है, तो $\left|\hat{u} - \vec{v}\right|^2$ बराबर है :

(1) $\frac{5}{2}$ (2) 7 (3) $\frac{11}{2}$ (4) 9

Ans. A

Sol. Unit vector $\hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{p}_1 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k},\ \vec{p}_2 = \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$$

$$\vec{p}_3 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

Now angle between $\hat{u}$ and $\vec{p}_1 = \frac{\pi}{2}$

$$\hat{u}\cdot\vec{p}_1 = 0 \Rightarrow \frac{x}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0$$

$$\Rightarrow x + z = 0$$

Now angle between $\hat{u}$ and $\vec{p}_2 = \frac{\pi}{3}$

$$\hat{u}\cdot\vec{p}_2 = \left|\hat{u}\right|\cdot\left|\vec{p}_2\right|\cos\frac{\pi}{3}$$

$$\Rightarrow \frac{y}{\sqrt{2}}+\frac{z}{\sqrt{2}}=\frac{1}{2}\Rightarrow y+z=\frac{1}{\sqrt{2}}$$

Now angle between $\hat{u}$ and $\vec{p}_3=\frac{2\pi}{3}$

$$\hat{u}\cdot\vec{p}_3=|\hat{u}|\cdot|\vec{p}_3|\cos\frac{2\pi}{3}$$

$$\Rightarrow \frac{x}{\sqrt{2}}+\frac{4}{\sqrt{2}}=\frac{-1}{2}$$

$$\Rightarrow x+y=\frac{-1}{\sqrt{2}}$$

from equation (i), (ii) and (iii) we get

$$x=\frac{-1}{\sqrt{2}}\quad y=0\quad z=\frac{1}{\sqrt{2}}$$

Thus $\hat{u}-\vec{v}=\frac{-1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}-\frac{1}{\sqrt{2}}\hat{i}-\frac{1}{\sqrt{2}}\hat{j}-\frac{1}{\sqrt{2}}\hat{k}n$

$$\hat{u}-\vec{v}=\frac{-2}{\sqrt{2}}\hat{i}-\frac{1}{\sqrt{2}}\hat{j}$$

$$\therefore|\hat{u}-\vec{v}|^2=\left(\sqrt{\frac{4}{2}+\frac{1}{2}}\right)^2=\frac{5}{2}$$

48. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ be two vectors such that $|\vec{a}|=1$, $\vec{a}.\vec{b}=2$ and $|\vec{b}|=4$.

If $\vec{c}=2\left(\vec{a}\times\vec{b}\right)-3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to :

माना दो सदिशों $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ तथा $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ के लिए $|\vec{a}|=1$, $\vec{a}.\vec{b}=2$ तथा $|\vec{b}|=4$ हैं। यदि $\vec{c}=2\left(\vec{a}\times\vec{b}\right)-3\vec{b}$ है, तो $\vec{b}$ तथा $\vec{c}$ के बीच कोण है :

(1) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (2) $\cos^{-1}\left(\frac{2}{3}\right)$ (3) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ (4) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Ans. C

Sol. Given $|\vec{a}|=1,|\vec{b}|=4,\vec{a}\cdot\vec{b}=2$

$$\vec{c}=2\left(\vec{a}\times\vec{b}\right)-3\vec{b}$$

Dot product with $\vec{a}$ on both sides

$$\vec{c}\cdot\vec{a}=-6 \qquad ...(1)$$

Dot product with $\vec{b}$ on both sides

$\vec{b}\cdot\vec{c} = -48$ ...(2)

$\vec{c}\cdot\vec{c} = 4\left|\vec{a}\times\vec{b}\right| + 9\left|\vec{b}\right|^2$

$\left|\vec{c}\right|^2 = 4\left[\left|a\right|^2\left|b\right|^2 - \left(a\cdot\vec{b}\right)^2\right] + 9\left|\vec{b}\right|^2$

$\left|\vec{c}\right|^2 = 4\left[(1)(4)^2 - (4)\right] + 9(16)$

$\left|\vec{c}\right|^2 = 4[12] + 144$

$\left|\vec{c}\right|^2 = 192$

$\therefore \cos\theta = \dfrac{\vec{b}\cdot\vec{c}}{\left|\vec{b}\right|\left|\vec{c}\right|}$

$\therefore \cos\theta = \dfrac{-48}{\sqrt{192}\cdot 4}$

$\therefore \cos\theta = \dfrac{-48}{8\sqrt{3}\cdot 4}$

$\therefore \cos\theta = \dfrac{-3}{2\sqrt{3}}$

$\therefore \cos\theta = \dfrac{-\sqrt{3}}{2} \Rightarrow \theta = \cos^{-1}\left(\dfrac{-\sqrt{3}}{2}\right)$

49. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{b}\right| = 1$ and $\left|\vec{b}\times\vec{a}\right| = 2$. Then $\left|\left(\vec{b}\times\vec{a}\right) - \vec{b}\right|^2$ is equal to :

माना दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार हैं कि $\left|\vec{b}\right| = 1$ तथा $\left|\vec{b}\times\vec{a}\right| = 2$ हैं। तो $\left|\left(\vec{b}\times\vec{a}\right) - \vec{b}\right|^2$ बराबर है :

(1) 3 (2) 1 (3) 5 (4) 4

Ans. C

Sol. $= \left|\vec{b}\times\vec{a}\right|^2 + \left|\vec{b}\right|^2 - 2\left(\vec{b}\times\vec{a}\right)\cdot\vec{b}$

$= 4 + 1 - 0 = 5$

50. Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}, \alpha, \beta \in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\dfrac{\pi}{4}$ and $\left|\vec{b}\right|^2 = 6$. If

$\vec{a}\cdot\vec{b}=3\sqrt{2}$, then the value of $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ is :

माना $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k},\alpha,\beta\in R$ है। माना एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a}$ तथा $\vec{b}$ के बीच कोण $\frac{\pi}{4}$ है तथा $\left|\vec{b}\right|^2=6$ है। यदि $\vec{a}\cdot\vec{b}=3\sqrt{2}$ है, तो $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ बराबर है :

(1) 75 (2) 95 (3) 85 (4) 90

Ans. D

Sol. $\vec{a}\cdot\vec{b}=3\sqrt{2}$

$$\left|\vec{a}\right|\left|\vec{b}\right|\cos\frac{\pi}{4}=3\sqrt{2}$$

$$\left|\vec{a}\right|\sqrt{6}\cdot\frac{1}{\sqrt{2}}=3\sqrt{2}$$

$$\left|\vec{a}\right|=\sqrt{6}=\sqrt{1+\alpha^2+\beta^2}$$

$$\alpha^2+\beta^2=5$$

$$\left|\vec{a}\times\vec{b}\right|^2=\left|\vec{a}\right|^2\left|\vec{b}\right|^2\sin^2\frac{\pi}{4}$$

$$=6\times6\times\frac{1}{2}=18$$

So, $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2=5\times18=90$

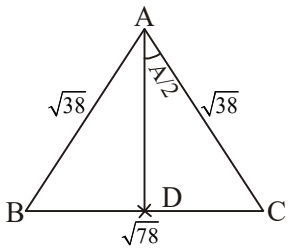
51. Consider a ΔABC where A(1, 3, 2), B(–2, 8, 0) and C(3, 6, 7). If the angle bisector of $\angle BAC$ meets the line BC at D, then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is :

एक ΔABC, जहाँ A(1, 3, 2), B(–2, 8, 0) तथा C(3, 6, 7) हैं, का विचार कीजिए। यदि कोण BAC की समद्विभाजक रेखा, BC को D पर मिलती है, तो सदिश $\overrightarrow{AD}$ के सदिश $\overrightarrow{AC}$ पर प्रक्षेप की लम्बाई है :

(1) $\frac{37}{2\sqrt{38}}$ (2) $\sqrt{19}$ (3) $\frac{\sqrt{38}}{2}$ (4) $\frac{39}{2\sqrt{38}}$

Ans. A

Sol.



Length of projection of $\overrightarrow{AD}$ a long $\overrightarrow{AC}$ $= AD\cos\frac{A}{2}$

$$= \left(\frac{2bc}{b+c}\right)\cos^2\frac{A}{2}$$

$$= \left(\frac{2bc}{b+c}\right)\left(\frac{1+\cos A}{2}\right)$$

$$= \frac{37}{2\sqrt{38}}$$

Method -II

$$D\left(\frac{1}{2}, 7, \frac{7}{2}\right)$$

$$\Rightarrow \overrightarrow{AD} = \frac{\hat{i}-8\hat{j}-3\hat{k}}{2}$$

$$\text{Length} = \left|\frac{\overrightarrow{AD}\,.\,\overrightarrow{AC}}{\left|\overrightarrow{AC}\right|}\right| = \frac{37}{2\sqrt{38}}$$

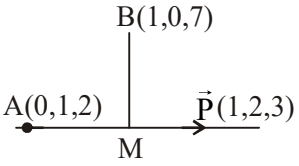
52. Let the image of the point (1,0,7) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ be the point (α, β, γ) Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

माना रेखा $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ में बिन्दु (1,0,7) का प्रतिबिम्ब (α, β, γ) है। तो निम्न में से कौनसा बिंदु, बिंदु (α, β, γ) से होकर जाने वाली, y और z-अक्षों से क्रमशः $\frac{2\pi}{3}$ तथा $\frac{3\pi}{4}$ का कोण बनाने वाली तथा x-अक्ष से न्यून कोण बनाने वाली रेखा पर है?

(1) $\left(1, 2, 1-\sqrt{2}\right)$ (2) $\left(3, -4, 3+2\sqrt{2}\right)$ (3) $\left(1, -2, 1+\sqrt{2}\right)$ (4) $\left(3, 4, 3-2\sqrt{2}\right)$

Ans. D

Sol.



$M = (\lambda, 1 + 2\lambda, 2 + 3\lambda)$

$\overrightarrow{BM} = (\lambda - 1, 2\lambda + 1, 3\lambda - 5)$

$\vec{P} = (1, 2, 3)$

$\overrightarrow{BM} \cdot \vec{P} = 0$

$1(\lambda - 1) + 2(2\lambda + 1) + 3(3\lambda - 5) = 0$

$\lambda = 1$

$M = (1, 3, 5)$

Image $(1, 6, 3)$

$m = \cos 2\pi / 3 = -1/2$

$n = \cos 3\pi / 4 = -\frac{1}{\sqrt{2}}$

$m^2 + n^2 + \ell^2 = 1$

$\ell^2 = \frac{1}{4}$

$\ell = \pm\frac{1}{2}$

So, required line is

$$\frac{x-1}{\pm\frac{1}{2}} = \frac{y-6}{-\frac{1}{2}} = \frac{z-3}{-\frac{1}{\sqrt{2}}}$$

$$\frac{x-1}{\pm 1} = \frac{y-6}{1} = \frac{z-3}{\sqrt{2}}$$

53. Let PQR be a triangle with R (–1, 4, 2). Suppose M (2, 1, 2) is the mid point of PQ. The distance of the centroid of ΔPQR from the point of intersection of the lines $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ and $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ is :

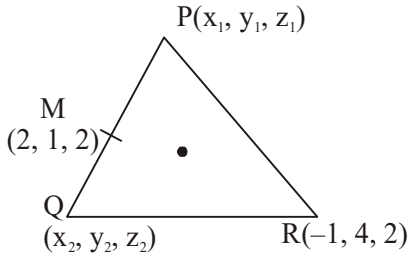
माना एक त्रिभुज PQR में R (–1, 4, 2) है। माना PQ का मध्य बिन्दु M (2, 1, 2) है। तो रेखाओं $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ तथा

$\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु से ΔPQR के केन्द्रक की दूरी है :

(1) $\sqrt{99}$ (2) $\sqrt{69}$ (3) 69 (4) 9

Ans. B

Sol.



Given two lines are

$\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}=\lambda$ (let)

$\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}=\mu$ (let)

for point of intersection

$x = 2, y = 2\lambda, z = -\lambda - 3$ (General point of line 1)

$x = \mu + 1, y = -3\mu - 3, z = \mu - 1$ (General point of line 2)

$2 = \mu + 1 \Rightarrow \mu = 1$

$2\lambda = -3\mu - 3 \Rightarrow \lambda = -3$

Point of intersection of two lines is $(2, -6, 0)$

$\because$ M is the mid point of PQ,

$\therefore x_1 + x_2 = 4, y_1 + y_2 = 2, z_1 + z_2 = 4$

Centroid of ΔPQR

$$=\left(\frac{x_1+x_2-1}{3},\frac{y_1+y_2+4}{3},\frac{z_1+z_2+2}{3}\right)$$

$= (1, 2, 2)$

$\therefore$ Distance between $(1, 2, 2)$ and $(2, -6, 0)$

$=\sqrt{1^2+8^2+2^2}=\sqrt{69}$

54. Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point R(1, 2, 3). If the centroid of the triangle PQR is (α, β, γ), then $\alpha^2 + \beta^2 + \gamma^2$ is :

माना रेखा $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ पर बिंदु R(1, 2, 3) से 6 इकाई दूरी पर दो बिंदु P तथा Q हैं। यदि त्रिभुज PQR का केन्द्रक $(\alpha,$

β, γ) है, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है :

(1) 36 (2) 18 (3) 24 (4) 26

Ans. B

Sol. Line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$

$P(8\lambda - 3, 2\lambda + 4, 2\lambda - 1)$

$PR = 6$

$\Rightarrow \lambda = 0, 1$

$\Rightarrow P(-3, 4, -1)$ Q(5, 6, 1)

Controid of $\Delta PQR = (1, 4, 1)$

$\alpha = 1,\ \beta = 4,\ \gamma = 1$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 18$

55. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 3\hat{j} + 3\hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 4$. Then $\left|\left(\vec{a} \times \vec{d}\right)\right|^2$ is equal to _____.

माना $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 3\hat{j} + 3\hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ हैं और एक सदिश $\vec{d}$ के लिए $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ और $\vec{a} \cdot \vec{d} = 4$ हैं। तो $\left|\left(\vec{a} \times \vec{d}\right)\right|^2$ बराबर है _____.

Ans. 128

Sol. $\left(\vec{b} - \vec{c}\right) \times \vec{d} = 0$

$\vec{b} - \vec{c} = \lambda \vec{d}$

Now taking dot with $\vec{a}$

$\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = \lambda \vec{a} \cdot \vec{d}$

$\Rightarrow 0 - 2 = \lambda 4 \Rightarrow \lambda = -\frac{1}{2}$

$\vec{b} = \vec{c} + \lambda \vec{d}$

$\vec{d} = 2\left(\vec{c} - \vec{b}\right) = 2\left(-\hat{i} + 2\hat{j} + \hat{k}\right),$

$\Rightarrow \left|\vec{d}\right|^2 = 4 \times 6$

$\Rightarrow \left|\vec{a}\right|^2 = 6$

$\left|\vec{a} \times \vec{d}\right|^2 = \left|\vec{a}\right|^2 \left|\vec{d}\right|^2 - \left(\vec{a} \cdot \vec{d}\right)^2$

$= 4 \times 6 \times 6 - 16 = 16(9 - 1) = 16 \times 8 = 128$

56. Let the three sides of a triangle ABC be given by the vectors $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$. Let G be the centroid of the triangle ABC. Then $6\left(|\overrightarrow{AG}|^2+|\overrightarrow{BG}|^2+|\overrightarrow{CG}|^2\right)$ is equal to_____.

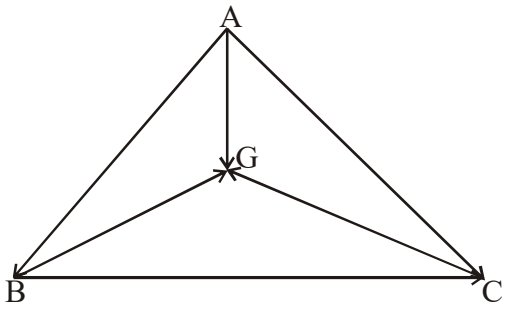
माना एक त्रिभुज ABC की तीन भुजाएँ सदिशों $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ और $3\hat{i}-4\hat{j}-4\hat{k}$ द्वारा दी गई हैं। माना त्रिभुज ABC का केन्द्रक G है। तो $6\left(|\overrightarrow{AG}|^2+|\overrightarrow{BG}|^2+|\overrightarrow{CG}|^2\right)$ बराबर है _____.

Ans. 164

Sol. $\overrightarrow{AB}=(2,-1,1)$

$\overrightarrow{BC}=(1,-3,-5)$

$\overrightarrow{AC}=(3,-4,-4)$



$\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$

Let A(0,0,0)

then B(2,–1,1), C(3,–4,–4)

$G=\left(\frac{5}{3},-\frac{5}{3},-1\right)$

$\left|\overrightarrow{AG}\right|^2=\frac{59}{9}$

$\left|\overrightarrow{BG}\right|^2=\frac{41}{9}$

$\left|\overrightarrow{CG}\right|^2=\frac{146}{9}$

$6\left\{\left|\overrightarrow{AG}\right|^2+\left|\overrightarrow{BG}\right|+\left|\overrightarrow{CG}\right|^2\right\}$

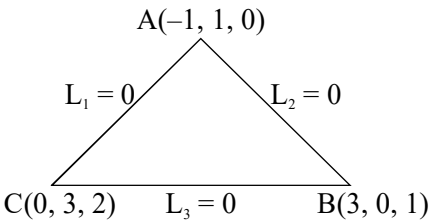
= 164

57. Let the area of the triangle formed by the lines $x+2=y-1=z$, $\frac{x-3}{5}=\frac{y}{-1}=\frac{z-1}{1}$ and $\frac{x}{-3}=\frac{y-3}{3}=\frac{z-2}{1}$ be A. Then A^2 is equal to _____.

माना रेखाओं $x + 2 = y - 1 = z$, $\frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1}$ और $\frac{x}{-3} = \frac{y-3}{3} = \frac{z-2}{1}$ द्वारा बने त्रिभुज का क्षेत्रफल A है। तो A^2 बराबर है _____.

Ans. 56

Sol.



$$L_1 : \frac{x+2}{1} = \frac{y-1}{1} = \frac{z-0}{1} \; : \; L_2 : \frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1} \; : \; L_3 : \frac{x-0}{-3} = \frac{y-3}{3} = \frac{z-2}{1}$$

$$A = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = \left|\frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -1 & 1 \\ 2 & 2 & 2 \end{vmatrix}\right|$$

$$A = \frac{1}{2}\left|\left(-4\hat{i} - 8\hat{j} + 12\hat{k}\right)\right|$$

$$A = \sqrt{4 + 16 + 36}$$

58. Let $\vec{c}$ be the projection vector $\vec{b} = \lambda\hat{i} + 4\hat{k}$, $\lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

माना कि $\vec{c}$, सदिश $\vec{b} = \lambda\hat{i} + 4\hat{k}$, $\lambda > 0$, का सदिश $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ पर प्रक्षेप है। यदि $|\vec{a} + \vec{c}| = 7$ है, तो सदिशों $\vec{b}$ और $\vec{c}$ द्वारा बनाए गए समान्तर चतुर्भुज का क्षेत्रफल है _____.

Ans. 16

Sol. $\vec{c} = \left(\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|}\right)\frac{\vec{a}}{|\vec{a}|}$

$$= \left(\frac{\lambda + 8}{9}\right)\left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$$

$|\vec{a} + \vec{c}| - = 7 \Rightarrow \lambda = 4$

Area of parallelogram

$$= \left|\vec{b}\times\vec{c}\right| = \left|\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{4}{3} & \frac{8}{3} & \frac{8}{3} \\ 4 & 0 & 4 \end{vmatrix}\right|$$

$= 16$

59. Let $\vec{a} = \hat{i}+\hat{j}+\hat{k}, \vec{b} = 2\hat{i}+2\hat{j}+\hat{k}$ and $\vec{d} = \vec{a}\times\vec{b}$. If $\vec{c}$ is a vector such that $\vec{a}\cdot\vec{c} = \left|\vec{c}\right|, \left|\vec{c}-2\vec{a}\right|^2 = 8$ and the angle between $\vec{d}$ and $\vec{c}$ is $\frac{\pi}{4}$, then $\left|10-3\vec{b}\cdot\vec{c}\right| + \left|\vec{d}\times\vec{c}\right|^2$ is equal to _____.

माना कि $\vec{a} = \hat{i}+\hat{j}+\hat{k}, \vec{b} = 2\hat{i}+2\hat{j}+\hat{k}$ और $\vec{d} = \vec{a}\times\vec{b}$ है। यदि $\vec{c}$ एक सदिश इस प्रकार है कि $\vec{a}\cdot\vec{c} = \left|\vec{c}\right|, \left|\vec{c}-2\vec{a}\right|^2 = 8$ और $\vec{d}$ और $\vec{c}$ के बीच का कोण $\frac{\pi}{4}$ है, तो $\left|10-3\vec{b}\cdot\vec{c}\right| + \left|\vec{d}\times\vec{c}\right|^2$ बराबर है _____.

Ans. 6

Sol. $\vec{a} = \hat{i}+\hat{j}+\hat{k}$

$\vec{b} = 2\hat{i}+2\hat{j}+\hat{k}$

$\vec{d} = \vec{a}\times\vec{b}$

$= -\hat{i}+\hat{j}$

$\left|\vec{c}-2\vec{a}\right|^2 = 8$

$|c|^2 + 4|a|^2 - 4(a.c) = 8$

$c^2 + 12 - 4c = 8$

$c^2 - 4c + 4 = 0$

$|c| = 2$

$\vec{d} = \vec{a}\times\vec{b}$

$\vec{d}\times\vec{c} = \left(\vec{a}\times\vec{b}\right)\times\vec{c}$

Now,

$$\left(|a||c|\sin\frac{\pi}{4}\right)^2 = \left((a.c).b - (b.c).a\right)^2$$

$4 = 4b^2 + (b.c)2(a^2) - 2(b.c)\,(a.b)$

$4 = 36 + 3x^2 - 20x$

Let $b.c = x$

$3x^2 - 20x + 32 = 0$

$3x^2 - 12x - 8x + 32 = 0$

$x = \frac{8}{3}, 4$; $b.c = \frac{8}{3}, 4$; $b.c = \frac{8}{3}$

Now, $|10 - 3b.c| + |d \times c|^2$; $|10 - 8| + (2)^2$

$\Rightarrow 6$

60. Let α, β $(\alpha \neq \beta)$ be the values of m, for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}\left(n^{\alpha} + n^{\beta}\right)$ is equal to :

माना कि α, β $(\alpha \neq \beta)$, m के वे मान है, जिनके लिए समीकरणों $x + y + z = 1$; $x + 2y + 4z = m$ और $x + 4y + 10z = m^2$ के अनन्त हल हैं। तब $\sum_{n=1}^{10}\left(n^{\alpha} + n^{\beta}\right)$ का मान बराबर है :

(1) 440 (2) 3080 (3) 3410 (4) 560

Ans. A

Sol. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix} = 1(20 - 16) - 1(10 - 4) + 1(4 - 2)$

$= 4 - 6 + 2 = 0$

For infinite solutions

$\Delta_x = \Delta_y = \Delta_z = 0$

$m^2 - 3x + 2 = 0$

$m = 1, 2$

$\alpha = 1, \beta = 2$

$\therefore \sum_{n=1}^{10}\left(n^{\alpha} + n^{\beta}\right) = \sum_{n=1}^{10} n^1 + \sum_{n=1}^{10} n^2$

$= \frac{10(11)}{2} + \frac{10(11)(21)}{6}$

$= 55 + 385$

$= 440$

61. Let $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ and $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha}$, $\alpha \in R$ be two lines, which intersect at the point B. If P is the foot of perpendicular from the point A(1, 1, –1) on L_2, then the value of $26\,\alpha(PB)^2$ is _____.

माना कि $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ और $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha}$, $\alpha \in R$, दो रेखाएँ हैं, जो कि बिन्दु B पर प्रतिच्छेद करती हैं।

यदि P, बिन्दु A(1, 1, –1) से L_2 पर लम्ब का पाद है, तो 26 $\alpha(PB)^2$ का मान है _____.

Ans. 216

Sol. Point B

$(3\lambda + 1, -\lambda + 1, -1) \equiv (2\mu + 2, 0, \alpha\mu - 4)$

$3\lambda + 1 = 2\lambda + 2$

$-\lambda + 1 = 0$

$-1 = \alpha\mu - 4$

$\lambda = 1, \mu = 1, \alpha = 3$

B(4, 0, –1)

Let Point 'P' is $(2\delta + 2, 0, 3\delta - 4)$

Dr's of AP $< 2\delta + 1, -1, 3\delta - 3>$

$$AP \perp L_2 \Rightarrow \delta = \frac{7}{13}$$

$$P\left(\frac{40}{13}, 0, \frac{-31}{13}\right)$$

$$2\sigma\delta(PB)^2 = 26 \times 3 \times \left(\frac{144}{169} + \frac{324}{169}\right)$$

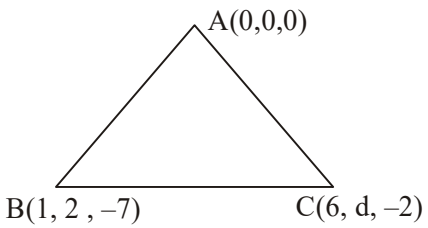
$= 216$

62. Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{AB} = \hat{i} + 2\hat{j} - 7\hat{k}$, $\overrightarrow{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\overrightarrow{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}$, $d > 0$. Then the square of the length of the largest side of the triangle ABC is _____.

माना एक त्रिभुज ABC का क्षेत्रफल $15\sqrt{2}$ है तथा सदिश $\overrightarrow{AB} = \hat{i} + 2\hat{j} - 7\hat{k}$, $\overrightarrow{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ और $\overrightarrow{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}$, $d > 0$ हैं। तो त्रिभुज ABC की सबसे बड़ी भुजा की लम्बाई का वर्ग है _____

Ans. 54

Sol.



$$\frac{1}{2}\begin{vmatrix} i & j & k \\ 1 & 2 & -7 \\ 6 & d & -2 \end{vmatrix} = \pm 30\sqrt{2}$$

$\hat{i}(-4+7d)-\hat{j}(-2+42)+\hat{k}(d-12)=\pm 30\sqrt{2}$

$\Rightarrow (7d-4)^2+40^2+(d-12)^2=1800$

$49d^2+16-56d+1600+d^2+144-24d=1800$

$50d^2-80d-40=0$

$5d^2-8d-4=0$

$5d^2-10d+2d-4=0$

$(5d+2)(d-2)=0$

$d=2$

$AB=\sqrt{54}$

$AC=\sqrt{36+4+4}=\sqrt{44}$

$BC=\sqrt{25+0+25}$

63. Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k},\vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$. If $\vec{a}\cdot\vec{c}=13$, then $\left(24-\vec{b}\cdot\vec{c}\right)$ is equal to _____.

माना $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k},\vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार है कि $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$ है। यदि $\vec{a}\cdot\vec{c}=13$, तो $\left(24-\vec{b}\cdot\vec{c}\right)$ बराबर है _____

Ans. 46

Sol. $\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}=(1,8,13)$

$\vec{a}\times(\vec{a}\times\vec{b})+\vec{a}\times(\vec{a}\times\vec{c})+\vec{a}\times(\vec{b}\times\vec{c})$

$=\vec{a}\times(\hat{i}+8\hat{j}+13k)$

$(\vec{a}\cdot\vec{b})\vec{a}-a^2\vec{b}+(\vec{a}\cdot\vec{c})\vec{a}-a^2\vec{c}+(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}=\vec{a}\times(\hat{i}+8\hat{j}+13\hat{k})$

$\Rightarrow -26\vec{a}-29\vec{b}+13\vec{a}-29\vec{c}+13\vec{b}+26\vec{c}=\vec{a}\times(\hat{i}+8\hat{j}+13\hat{k})$

$\Rightarrow \quad -13\vec{a}-16\vec{b}-3\vec{c}=\vec{a}\times(\hat{i}+8\hat{j}+13\hat{k})$

$\Rightarrow \quad -13\vec{a}\cdot\vec{b}-16b^2-3\vec{b}\cdot\vec{c}=\vec{a}\times(\hat{i}+8\hat{j}+13\hat{k})\cdot\vec{b}$

$\Rightarrow \quad (-13)(-26)-16(50)-3\vec{b}\cdot\vec{c}=\begin{vmatrix}2 & -3 & 4\\ 1 & 8 & 13\\ 3 & 4 & -5\end{vmatrix}$

$\Rightarrow -462 - 3\vec{b}\cdot\vec{c} = -396$

$\Rightarrow \vec{b}\cdot\vec{c} = -22$

Hence $24 - \vec{b}\cdot\vec{c} = 46$

64. Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a} = (\vec{b}+\vec{c})\times\vec{a}$ and $\vec{r}\cdot(\vec{b}-\vec{c}) = 0$, then $\dfrac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____.

माना $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ तथा $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ तीन सदिश दिये गये है यदि $\vec{r}$ एक सदिश इस प्रकार है कि $\vec{r}\times\vec{a} = (\vec{b}+\vec{c})\times\vec{a}$ तथा $\vec{r}\cdot(\vec{b}-\vec{c}) = 0$ है, तो $\dfrac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ बराबर होगा _____

Ans. 569

Sol. $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$

$\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$

$\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$

$\vec{r}\times\vec{a} = (\vec{b}+\vec{c})\times\vec{a}$

$(\vec{r} - (\vec{b}+\vec{c}))\times\vec{a} = 0$

$\Rightarrow \vec{r} = (\vec{b}+\vec{c}) + \lambda\vec{a}$

$\vec{r} = (20\hat{i} + 5\hat{j} - 12\hat{k}) + \lambda(9\hat{i} - 13\hat{j} + 25\hat{k})$

$= (20 + 9\lambda)\hat{i} + (5 - 13\lambda)\hat{j} + (25\lambda - 12)\hat{k}$

Now, $\vec{r}\cdot(\vec{b}-\vec{c}) = 0$

$\vec{r}\cdot(-14\hat{i} + 9\hat{j} - 14\hat{k}) = 0$

Now $-14(20 + 9\lambda) + 9(5 - 13\lambda) - 14(25\lambda - 12) = 0$

$-593\lambda - 67 = 0$

$\lambda = -\dfrac{67}{593}$

$\therefore \vec{r} = (\vec{b}+\vec{c}) - \dfrac{67}{593}\vec{a}$

$$\frac{\left|593\vec{r}+67\vec{a}\right|^2}{\left|593\right|^2}=\left|\vec{b}+\vec{c}\right|^2=\left|20\hat{i}+5\hat{j}-12\hat{k}\right|^2$$

$= 569$

65. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}|=1$, $|\vec{b}|=4$, and $\vec{a}\cdot\vec{b}=2$. If $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α, then $192\sin^2\alpha$ is equal to _____.

माना दो सदिशों $\vec{a}$ तथा $\vec{b}$ के लिए $|\vec{a}|=1$, $|\vec{b}|=4$, तथा $\vec{a}\cdot\vec{b}=2$ हैं। यदि $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ है तथा $\vec{b}$ और $\vec{c}$ है के बीच कोण α है, तो $192\sin^2\alpha$ बराबर है _____.

Ans. 48

Sol. $\vec{b}\cdot\vec{c}=\left(2\vec{a}\times\vec{b}\right)\cdot\vec{b}-3|b|^2$

$|b||c|\cos\alpha=-3|b|^2$

$|c|\cos\alpha=-12,\ \text{as}\ |b|=4$

$\vec{a}\cdot\vec{b}=2$

$\cos\theta=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$

$|c|^2=\left|\left(2\vec{a}\times\vec{b}\right)-3\vec{b}\right|^2$

$=64\times\frac{3}{4}+144=192$

$|c|^2\cos^2\alpha=144$

$192\cos^2\alpha=144$

$192\sin^2\alpha=48$

66. Let $\vec{a}=3\hat{i}+2\hat{j}+\hat{k},\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ and $\vec{c}$ be a vector such that $\left(\vec{a}+\vec{b}\right)\times\vec{c}=2\left(\vec{a}\times\vec{b}\right)+24\hat{j}-6\hat{k}$ and $\left(\vec{a}-\vec{b}+\hat{i}\right).\vec{c}=-3$. Then $|\vec{c}|^2$ is equal to _____.

माना $\vec{a}=3\hat{i}+2\hat{j}+\hat{k},\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार हैं कि $\left(\vec{a}+\vec{b}\right)\times\vec{c}=2\left(\vec{a}\times\vec{b}\right)+24\hat{j}-6\hat{k}$ तथा $\left(\vec{a}-\vec{b}+\hat{i}\right).\vec{c}=-3$ हैं। तो $|\vec{c}|^2$ बराबर है _____

Ans. 38

Sol. $(\vec{a}+\vec{b})\times\vec{c}=2(\vec{a}\times\vec{b})+24\hat{j}-6\hat{k}$

$(5\hat{i}+\hat{j}+4\hat{k})\times\vec{c}=2(7\hat{i}-7\hat{j}-7\hat{k})+24\hat{j}-6\hat{k}$

$$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\5&1&4\\x&y&z\end{vmatrix}=14\hat{i}+10\hat{j}-20\hat{k}$$

$\Rightarrow \hat{i}(z-4y)-\hat{j}(5z-4x)+\hat{k}(5y-x)=14\hat{i}+10\hat{j}-20\hat{k}$

$z-4y=14, 4x-5z=10, 5y-x=-20$

$(a-b+i)\cdot\vec{c}=-3$

$(2\hat{i}+3\hat{j}-2\hat{k})\cdot\vec{c}=-3$

$2x+3y-2z=-3$

$\therefore x=5, y=-3, z=2$

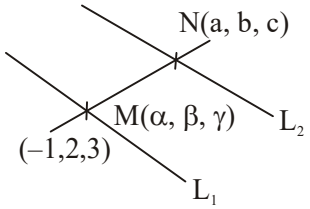
$|\vec{c}|^2=25+9+4=38$

67. Let a line passing through the point (–1, 2, 3) intersect the lines $L_1:\frac{x-1}{3}=\frac{y-2}{2}=\frac{z+1}{-2}$ at M(α, β, γ) and $L_2:\frac{x+2}{-3}=\frac{y-2}{-2}=\frac{z-1}{4}$ at N(a, b, c). Then the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equal _____.

माना बिन्दु (–1, 2, 3) से होकर जाने वाली रेखा, रेखा $L_1:\frac{x-1}{3}=\frac{y-2}{2}=\frac{z+1}{-2}$ को M(α, β, γ) तथा रेखा $L_2:\frac{x+2}{-3}=\frac{y-2}{-2}=\frac{z-1}{4}$ को N(a, b, c) पर काटती है। तो $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ बराबर है _____

Ans. 196

Sol.



M(3λ + 1, 2λ + 2, –2λ – 1)

N(–3μ – 2, – 2μ + 2, 4μ + 1)

$\frac{3\lambda+2}{-3\mu-1}=\frac{2\lambda}{-2\mu}=\frac{-2\lambda-4}{4\mu-2}\Rightarrow\mu=2,\lambda=4$

So, $\begin{matrix}\alpha+\beta+\gamma=14\\a+b+c=-1\end{matrix}\Rightarrow\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}=196$

68. Let Q and R be the feet of perpendiculars from the point (a,a,a) on the lines x = y, z = 1 and x = –y, z = –1

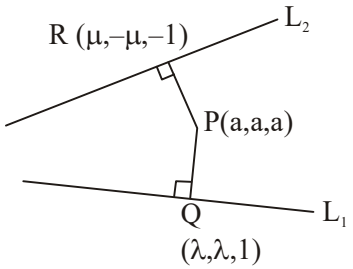
respectively. If $\angle QPR$ is a right angle, then $12a^2$ is equal to _____.

माना बिन्दु P(a,a,a) से रेखाओं x = y, z = 1 तथा x = –y, z = –1 पर डाले गए लंबों के पाद क्रमशः Q तथा R हैं। यदि $\angle QPR$ एक समकोण हैं, तो $12a^2$ बराबर है _____.

Ans. 12

Sol. $L_1 : \frac{x}{1} = \frac{y}{1} = \frac{z-1}{0} = \lambda$ (let)

$L_2 : \frac{x}{1} = \frac{y}{-1} = \frac{z+1}{0} = \mu$ (let)



$PQ \perp L_1$
$(\lambda - a)1 + (\lambda - a)1 + (1 - a)0 = 0$
$\lambda = a$
$PR \perp L_2$
$1(\mu - a) + 1(\mu + a) + 0(-1 - a) = 0$
$\mu - a + \mu + a = 0$
$\mu = 0$
$\therefore$ Point Q(a,a,1) and R(0,0,–1)

According to question

$\angle QPR = 90^o$

Direction ratio of PR = (a, a, a + 1)

Direction ratiom of PQ = (0, 0, a – 1)

$a \times 0 + 0 \times a + (a + 1)(a - 1) = 0$

$a^2 = 1$

$\Rightarrow 12a^2 = 12 \times 1 = 12$

69. A line passes through A (4, –6, –2) and B (16, –2, 4). Then point P (a, b, c) , where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points P(a, b, c) and Q(4, –12, 3) is equal to _____.

एक रेखा, बिन्दुओं A (4, –6, –2) तथा B (16, –2, 4) से होकर जाती है। रेखा AB पर बिन्दु P(a, b, c), जहाँ a, b, c ऋणोत्तर पूर्णांक हैं, बिन्दु A से 21 इकाई की दूरी पर है। बिन्दुओं P(a, b, c) तथा Q(4, –12, 3) के बीच दूरी है _____

Ans. 22

Sol. $A(4-6-2)$ $B(16-2,4)$

Equation of line AB :

$$\frac{x-4}{6}=\frac{y+6}{2}=\frac{z+2}{3}$$

$$\frac{x-4}{\frac{6}{7}}=\frac{y+6}{\frac{2}{7}}=\frac{z+2}{\frac{3}{7}}=21$$

$x=22,\quad y=0,\quad z=7$

$P(22,0,7),\quad Q(4,-12,3)$

$PQ=\sqrt{(18)^2+(12)^2+(4)^2}$

$=\sqrt{484}=22$